quadsim: A Modular, Validated Drone Reinforcement-Learning Research Framework on NVIDIA Isaac Lab

Technical Report

Drone_Sim_RL project

November 2026

Abstract

We present *quadsim*, a GPU-parallel quadrotor simulation and reinforcement-learning (RL) platform built on Isaac Lab v2.3.2 and Isaac Sim 5.1. The platform couples a framework-agnostic PyTorch model of multirotor physics (asymmetric first-order motors, rotor and body drag, gyroscopic and reaction torques, ground effect, wind) with PhysX rigid-body integration, models the onboard sensing chain (IMU, barometer, state estimate, depth/RGB cameras, range sensing) including noise, bias and latency, and exposes hover, trajectory tracking, waypoint and vision-based obstacle-avoidance tasks with deployable observations only. Every component is validated against analytic solutions or an independent integrator: PhysX trajectories match the reference integrator to 0.4 mm over 1.5 s of open-loop flight, RTX-rendered depth matches our analytic GPU ray caster to 7 mm median error, and the exported ONNX policy, driven by a NumPy deployment interface, reproduces simulator observations and actions to 10^{-6} in closed loop. PPO policies trained in 10–40 minutes on an 8 GB consumer GPU outperform classical cascade and geometric controllers under identical noise, latency and randomization (hover 99.9 vs 90.3 % success, tracking 12.6 vs 46.6 cm RMSE); a 360° range-sensing navigation policy reaches 84 % success in a dense forest, and a depth-camera policy trained on analytic depth transfers without significant degradation to RTX-rendered depth (64 vs 60 % success). Negative results (no significant success gain from domain randomization on hover, an under-trained SAC baseline) are reported alongside. Finally, the exported policies were flown as PX4 v1.16 offboard controllers in software-in-the-loop simulation (Gazebo x500) through a ROS 2 bridge: hover, tracking and waypoint policies transfer at the control level (hover 0.8 cm to 1.0 cm, tracking 2–3 cm RMS w.r.t. the onboard estimate, as in quadsim), while the cross-check exposed four bridge defects, a heave limit cycle caused by the thrust-map nonlinearity (removed by an exact thrust-curve map), PX4 EKF2's correlated estimate error, an allocation-geometry asymmetry of the rate loop, and a velocity-loop mismatch that limits the velocity-interface navigation policy (2/6 vs 87 % success; 4/6 after matching PX4's velocity loop). A first SITL-informed training distribution made the heave mode worse (negative result); a corrected one (3 seeds) removed the limit cycle in all 9 SITL flights at the cost of vertical damping and out-of-distribution robustness. Building on this validated core, quadsim was turned into a general research framework (Section 8): registries and plug-ins for vehicles (YAML files, any rotor count), control interfaces (motor-level to position control), algorithms and benchmarks; configuration terms for sensors (rate, latency, history), rewards, terminations, costs, randomization events and curricula; GPU-native SAC/TD3 with asymmetric critics; PPO-Lagrangian constrained RL; parameter-shared multi-agent training with a centralized critic; drone racing and language-conditioned navigation tasks; and an imitation-learning / vision-language-action data path (HDF5 demonstrations, behaviour cloning, DAgger, LeRobot v3.0 export, VLA adapters). A regression check shows the refactored core reproduces 59 reference metrics bit-exactly; the new SAC matches PPO on hover (100 %) where the earlier SAC failed; training a racing policy on both turning directions raises figure-8 completion from 4 to 73 % (3 seeds); the Lagrangian keeps a speed constraint around its budget, but one of three seeds never learns the task; and language-conditioned behaviour cloning reaches the expert's 100 % after two DAgger rounds, with 74 ± 15 % for unseen phrasings using a bag-of-words encoder and 97 ± 1 % with a frozen pretrained CLIP text encoder. A subsequent independent code review found and fixed about twenty defects — the most consequential a sampling bias of the evaluator that over-weighted short episodes — after which every result was re-measured (Section 8.4). We

report what is verified and, explicitly, what is not (no hardware flight yet), and give a prioritized roadmap.

Contents

1 Introduction and scope **4**

2 Framework selection and related work **4**

3 System architecture **4**

4 Mathematical models **5**
4.1 Frames and rigid-body dynamics . . . 5
4.2 Rotors and motors . . . 5
4.3 Aerodynamics . . . 5
4.4 Wind and disturbances . . . 6
4.5 Control allocation and classical controllers . . . 6
4.6 Sensor models . . . 6

5 Reinforcement-learning environments **6**
5.1 Observation and action design . . . 6
5.2 Tasks and rewards . . . 7
5.3 Domain randomization . . . 7
5.4 Learning algorithm . . . 7

6 Validation and experiments **7**
6.1 M1: simulator foundation . . . 7
6.2 M2: quadrotor dynamics . . . 8
6.3 M3: sensors, recording and visualization . . . 8
6.3.1 Upstream issue found and patched . . . 9
6.4 M4: state-based reinforcement learning . . . 9
6.5 M5: obstacle-avoidance navigation (range and depth) . . . 10
6.6 Domain-randomization ablation . . . 12
6.7 M6: export and deployment pipeline . . . 13
6.8 Off-policy baseline: SAC . . . 13
6.9 M7: throughput, memory and containerization . . . 13

7 PX4 SITL validation and sim-to-sim analysis **13**
7.1 Setup and deployment pipeline . . . 14
7.2 Defects found by SITL . . . 15
7.3 Estimator . . . 15
7.4 Hover and set-point steps . . . 15
7.5 Training distributions v1 / v2 and multi-seed comparison . . . 17
7.6 Trajectory tracking and waypoints . . . 17
7.7 Obstacle navigation through the velocity interface . . . 18
7.8 Summary of the cross-check . . . 20

8 From platform to research framework **20**
8.1 Extension points . . . 20
8.2 Validation of the framework . . . 22
8.3 Findings . . . 22
8.4 Review, corrections and multi-seed replication . . . 26
8.5 Tools for VLA research and rigorous reporting . . . 28

9 Limitations **29**

10 Roadmap toward real-world deployment **30**

11 Reproducibility **30**

1 Introduction and scope

Learning-based quadrotor control has reached champion-level performance in drone racing [1] and robust low-level control learned in seconds [2]. Such results rest on simulators whose physics and sensing are close enough to reality, and on disciplined interfaces between policy and vehicle. The goal of this project is a *research-grade* platform with these properties, prioritizing verified functionality over feature count.

Hardware note. The original brief targeted an RTX 5070 (12 GB). The actual machine has an **RTX 5060 with 8 GB** (Blackwell, sm_120), Ubuntu 24.04, an i5-14600KF and 32 GB RAM. All budgets (env counts, image sizes) were chosen for 8 GB.

Validation policy. Each milestone has measurable acceptance criteria that are executed by scripts in `scripts/validation`; results are stored as JSON in `docs/results` and summarized in `docs/PROGRESS.md`. RL results distinguish *training execution* from *demonstrated policy performance*, which is always measured by a separate deterministic evaluation with confidence intervals.

2 Framework selection and related work

Isaac Lab offers GPU-parallel PhysX simulation, tiled RTX rendering and RL integrations. At the time of writing the latest releases were v2.3.2 (Isaac Sim 5.1) and v3.0.0 (Isaac Sim 6.2). We selected **v2.3.2 + Isaac Sim 5.1**: it is the latest stable pair, it is available as an official container (`nvcr.io/nvidia/isaac-lab:2.3.2`), and v3.0 introduces breaking changes (quaternion convention $wxyz \to xyzw$, multi-backend refactor). The physics core of quadsim is framework-agnostic to ease a later migration.

Existing quadrotor components. Isaac Lab's `Isaac-Quadcopter-Direct` task applies a body wrench directly from actions without motor dynamics; the `isaaclab_contrib` multirotor asset (from the Aerial Gym authors [4]) adds a first-order thruster model with an allocation matrix. Aerial Gym [4] and OmniDrones [5] provide broader multirotor suites on Isaac Gym / Isaac Sim. We reused ideas (asymmetric motor time constants, allocation matrices, depth-based navigation in cluttered environments) but implemented our own model to obtain (i) rotor drag, gyroscopic, reaction and ground-effect terms, (ii) unit-testability without a simulator, and (iii) an independent reference integrator for validation. The vehicle parameters follow the PX4 SITL Gazebo model of the Holybro X500 so that results can be cross-checked against PX4 SITL later.

Network constraint. NVIDIA's asset CDN (Nucleus/S3) was effectively unreachable from the development network (the stock quadcopter task did not finish loading its USD within 625 s), and NGC refused direct container pulls (HTTP 403). Consequently all assets are *procedural* and the container base image is fetched through the user's proxy by a small registry client with digest verification.

3 System architecture

The code base is layered (Fig. 1):

- `quadsim.core` (pure PyTorch): parameters, motor/aerodynamic model, reference integrators, controllers, sensor and disturbance models, analytic ray casting (PyTorch and NVIDIA Warp kernels).
- `quadsim.isaac`: procedural USD quadrotor (explicit `MassAPI`: mass, inertia and CoM are written from the parameters, not derived from geometry) and version-gated runtime patches for upstream issues.
- `quadsim.tasks`: Isaac Lab `DirectRLEnv` base environment and tasks, PPO configurations and classical baselines that act through the same action interface.

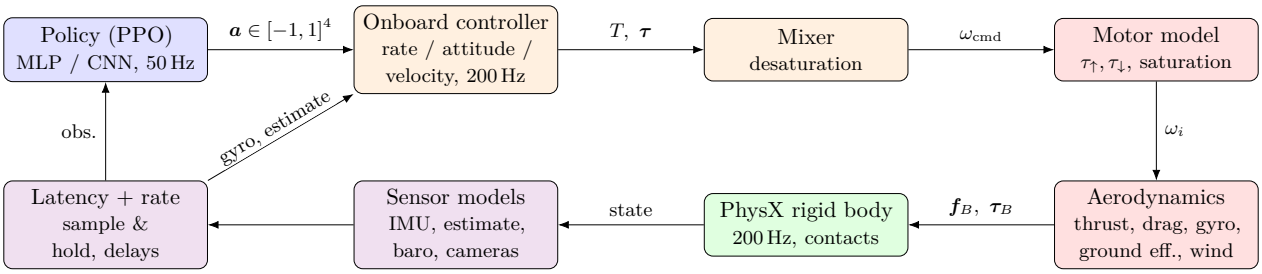


Figure 1: Per-step data flow. Orange, red and violet blocks belong to the framework-agnostic PyTorch core (`quadsim.core`) and are unit-tested without Isaac Sim; the green block is PhysX (Isaac Sim). Rendering (RTX cameras) is an alternative source of image observations.

- `quadsim.rl`, `quadsim.data`, `quadsim.deploy`: networks and policy I/O, HDF5 recording, export and the NumPy deployment interface.

Isaac Lab itself is not modified; it is installed from a pinned tag into `third_party/`.

Simulation timeline. Physics runs at 200 Hz, the policy at 50 Hz (decimation 4). Within each physics step: sensors are sampled; the delayed policy command is passed to the onboard controller (which uses the *noisy* gyro and state estimate); the mixer produces rotor-speed commands; motor states are advanced exactly; the aerodynamic wrench is evaluated and written to PhysX through Isaac Lab's wrench composer (body frame); PhysX integrates the rigid body and contacts.

4 Mathematical models

4.1 Frames and rigid-body dynamics

The world frame W is z-up (ENU-like), the body frame B is FLU, and $\boldsymbol{q} \in S^3$ (scalar first) rotates B into W with rotation matrix $\boldsymbol{R}(\boldsymbol{q})$. With mass m and principal inertia $\boldsymbol{J} = \mathrm{diag}(J_x, J_y, J_z)$,

$$\dot{\boldsymbol{p}} = \boldsymbol{v}, \qquad \dot{\boldsymbol{v}} = \tfrac{1}{m}\boldsymbol{R}\,\boldsymbol{f}_B + \boldsymbol{g} + \tfrac{1}{m}\boldsymbol{f}_W^{\text{ext}}, \tag{1}$$

$$\dot{\boldsymbol{q}} = \tfrac{1}{2}\,\boldsymbol{q} \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix}, \qquad \boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_B - \boldsymbol{\omega} \times \boldsymbol{J}\boldsymbol{\omega}, \tag{2}$$

with $\boldsymbol{g} = (0, 0, -9.81)$. PhysX integrates these equations with a symplectic (semi-implicit) Euler scheme and gyroscopic forces enabled; the reference integrator offers RK4 and the same semi-implicit scheme.

4.2 Rotors and motors

Rotor i at lever arm $\boldsymbol{r}_i$ (relative to the CoM) with spin $s_i = +1$ (CCW) or -1 spins at ω_i:

$$\dot{\omega}_i = \frac{\omega_{\text{cmd},i} - \omega_i}{\tau_i}, \qquad \tau_i = \begin{cases} \tau_\uparrow & \omega_{\text{cmd},i} \geq \omega_i \\ \tau_\downarrow & \text{otherwise} \end{cases}, \qquad \omega_{\text{cmd},i} \in [\omega_{\min}, \omega_{\max}], \tag{3}$$

discretized exactly under zero-order hold: $\omega_i^+ = \omega_{\text{cmd}} + (\omega_i - \omega_{\text{cmd}})e^{-\Delta t/\tau_i}$. Thrust and reaction torque are $T_i = k_f\,\omega_i^2\,g_e(h_i)$ and $Q_i = -s_i\,(k_m T_i + J_r\dot{\omega}_i)$, the latter including the spin-up reaction of the rotor inertia J_r.

4.3 Aerodynamics

With air-relative velocity $\boldsymbol{v}_{\text{rel}} = \boldsymbol{v}_B - \boldsymbol{w}_B$ (wind $\boldsymbol{w}$) and its component $\boldsymbol{v}_{\perp,i}$ orthogonal to the rotor axis at the hub ($\boldsymbol{v}_{\text{rel}} + \boldsymbol{\omega} \times \boldsymbol{r}_i$),

$$\boldsymbol{f}_{\text{rd},i} = -c_{\text{rd}}\,\omega_i\,\boldsymbol{v}_{\perp,i} \quad \text{(blade flapping / induced drag)}, \qquad \boldsymbol{m}_{\text{rm},i} = -c_{\text{rm}}\,s_i\,\omega_i\,\boldsymbol{v}_{\perp,i}, \tag{4}$$

$$\boldsymbol{f}_{\text{body}} = -\boldsymbol{D}_1\boldsymbol{v}_{\text{rel}} - \boldsymbol{D}_2\,|\boldsymbol{v}_{\text{rel}}| \odot \boldsymbol{v}_{\text{rel}}, \qquad \boldsymbol{\tau}_{\text{gyr}} = -\boldsymbol{\omega} \times \textstyle\sum_i J_r s_i \omega_i\,\boldsymbol{e}_z, \tag{5}$$

and the in-ground-effect gain of Cheeseman and Bennett, $g_e(h) = \left(1 - (R/4h)^2\right)^{-1}$, capped at $h \geq R/2$. The total body wrench is

$$\boldsymbol{f}_B = \sum_i \left(T_i \boldsymbol{e}_z + \boldsymbol{f}_{\mathrm{rd},i}\right) + \boldsymbol{f}_{\mathrm{body}}, \quad \boldsymbol{\tau}_B = \sum_i \left[\boldsymbol{r}_i \times \left(T_i \boldsymbol{e}_z + \boldsymbol{f}_{\mathrm{rd},i}\right) + \boldsymbol{m}_{\mathrm{rm},i} + Q_i \boldsymbol{e}_z\right] + \boldsymbol{\tau}_{\mathrm{gyr}}. \tag{6}$$

The X500 preset uses the PX4 Gazebo values ($k_f = 8.549{\times}10^{-6}$, $k_m = 0.016$, $\tau_\uparrow = 12.5\,\mathrm{ms}$, $\tau_\downarrow = 25\,\mathrm{ms}$, $\omega_{\max} = 1000\,\mathrm{rad/s}$, $c_{\mathrm{rd}} = 8.06 \times 10^{-5}$); its composite mass and inertia (base + rotors, parallel-axis theorem) are $m = 2.064\,\mathrm{kg}$, $\boldsymbol{J} = \mathrm{diag}(0.0239, 0.0239, 0.0440)\,\mathrm{kg\,m^2}$, hover speed $769.6\,\mathrm{rad/s}$ and thrust-to-weight 1.69. Body drag coefficients are engineering estimates (not identified).

4.4 Wind and disturbances

Wind is $\boldsymbol{w} = \bar{\boldsymbol{w}} + \boldsymbol{w}_{\mathrm{t}} + \boldsymbol{w}_{\mathrm{g}}$: a per-episode mean, Gauss–Markov turbulence $\boldsymbol{w}_{\mathrm{t}}^{+} = a\boldsymbol{w}_{\mathrm{t}} + \sigma\sqrt{1-a^2}\,\boldsymbol{\eta}$ with $a = e^{-\Delta t/\tau_w}$ (exact discretization of a Dryden-like first-order process with stationary standard deviation σ) and discrete $1-\cos$ gusts (MIL-F-8785C shape) with Poisson arrivals. Push disturbances apply random finite-duration forces.

4.5 Control allocation and classical controllers

The allocation $[F_z, \tau_x, \tau_y, \tau_z]^\top = \boldsymbol{A}\boldsymbol{T}$ with rows $(1,\, y_i,\, -x_i,\, -s_i k_m)$ is inverted with PX4-style sequential desaturation: roll/pitch are scaled only if their spread exceeds the actuator range, the collective is shifted to preserve roll/pitch authority, and yaw is added with the largest feasible scale. The onboard rate controller is $\boldsymbol{\tau} = \boldsymbol{J}(\boldsymbol{K}_p \boldsymbol{e}_\omega + \boldsymbol{K}_i \int \boldsymbol{e}_\omega - \boldsymbol{K}_d \dot{\boldsymbol{\omega}}_f) + \boldsymbol{\omega} \times \boldsymbol{J}\boldsymbol{\omega}$ (derivative on the filtered measurement), the attitude controller is a quaternion P law on the rotation vector of $\boldsymbol{q}^{-1} \otimes \boldsymbol{q}_{\mathrm{sp}}$, and the position controller is a P–PI cascade producing a thrust vector with acceleration and tilt limits. The geometric baseline uses Lee et al.'s SE(3) law [6] $\boldsymbol{f} = m(-\boldsymbol{K}_x \boldsymbol{e}_x - \boldsymbol{K}_v \boldsymbol{e}_v + g\boldsymbol{e}_3 + \ddot{\boldsymbol{p}}_d)$.

4.6 Sensor models

IMU: $\tilde{\boldsymbol{\omega}} = (\boldsymbol{I} + \boldsymbol{M}_g)\boldsymbol{\omega} + \boldsymbol{b}_g + \boldsymbol{n}_g$, $\tilde{\boldsymbol{a}} = (\boldsymbol{I} + \boldsymbol{M}_a)\boldsymbol{R}^\top(\dot{\boldsymbol{v}} - \boldsymbol{g}) + \boldsymbol{b}_a + \boldsymbol{n}_a$, where the specific force is obtained by finite differences of the PhysX velocity (exact for the symplectic scheme), white noise has density σ_c (discrete std $\sigma_c/\sqrt{\Delta t_s}$), and biases are a per-episode turn-on value plus a Gauss–Markov drift with random-walk intensity σ_{rw} (stationary std $\sigma_{rw}\sqrt{\tau/2}$). Default values are the PX4/rotors SITL IMU parameters. Measurements are sampled and held at the sensor rate and pass a per-episode random delay. **State estimate:** $\hat{\boldsymbol{p}} = \boldsymbol{p} + \boldsymbol{b}_p + \boldsymbol{x}_p + \boldsymbol{d}_p + \boldsymbol{n}_p$, $\hat{\boldsymbol{v}} = \boldsymbol{v} + \boldsymbol{x}_v + \boldsymbol{n}_v$, attitude with a multiplicative rotation-vector error, all delayed by a per-episode latency; $\boldsymbol{b}_p \sim \mathcal{N}(0, \mathrm{diag}\,\boldsymbol{\sigma}_b^2)$ is a per-flight bias, $\boldsymbol{x}_p, \boldsymbol{x}_v$ are first-order Gauss–Markov processes ($\dot{\boldsymbol{x}} = -\boldsymbol{x}/\tau + \sigma\sqrt{2/\tau}\,\boldsymbol{w}$, exact discretization $\boldsymbol{x}_{k+1} = a\boldsymbol{x}_k + \sigma\sqrt{1-a^2}\,\boldsymbol{\xi}_k$, $a = e^{-\Delta t/\tau}$), $\boldsymbol{d}_p$ a random walk and $\boldsymbol{n}$ white noise. The v0 tasks use only the white and random-walk terms (EKF/VIO/mocap-like); the parameters of the correlated terms of the v1 tasks were identified from PX4 EKF2 in SITL (Section 7.3). **Barometer:** ISA conversion with noise and Gauss–Markov bias. **Depth camera:** distance-to-image-plane with stereo-like noise $\sigma(z) = \sigma_0 + kz^2$, quantization, per-pixel dropouts, minimum/maximum range; no-return pixels read as maximum range. **Analytic ray casting:** exact ray–plane, ray–finite-cylinder (side and cap) and ray–oriented-box intersections, implemented in PyTorch and as an NVIDIA Warp kernel with $O(NR)$ memory.

5 Reinforcement-learning environments

5.1 Observation and action design

The `policy` observation contains only quantities available onboard: estimated body-frame velocity, the first two columns of the estimated rotation matrix (continuous attitude representation), gyro rates, the previous action and task information expressed in the estimated body frame. The `critic` additionally receives the true state, normalized rotor speeds, wind, disturbance force, and

normalized domain-randomization parameters (asymmetric actor-critic [7]). Four action interfaces are available: individual rotor thrusts, collective thrust and body rates (CTBR, the default for agile tasks following [3]), collective thrust and attitude, and velocity commands (default for navigation, matching PX4 offboard velocity control).

5.2 Tasks and rewards

All rewards are integrated over the policy step Δt; crashes (ground contact, excessive tilt, velocity, distance, or obstacle collision) terminate the episode with a penalty.

- **Hover/recovery**: initial states up to $\pm 1.5\,\mathrm{m}$ offset, 57° tilt, $1.5\,\mathrm{m/s}$, $3\,\mathrm{rad/s}$; reward $1 - \tanh(\|e_p\|/0.8) + 0.5\,e^{-(\|e_p\|/0.1)^2} + 0.2\cos^2(e_\psi/2)$ minus velocity, body-rate and action-rate penalties.
- **Tracking**: references $\boldsymbol{p}_d(t) = \boldsymbol{c} + \sum_k \boldsymbol{A}_k \sin(\boldsymbol{\omega}_k t + \boldsymbol{\phi}_k)$ covering circles, figure-eights and random smooth curves (peak speed $\leq 4\,\mathrm{m/s}$); the policy observes the reference at $t + \{0, 0.1, 0.2, 0.4, 0.6, 1.0\}\,\mathrm{s}$.
- **Waypoints**: five random waypoints (separation $\geq 1.5\,\mathrm{m}$); progress reward $\|\boldsymbol{p} - \boldsymbol{w}\|_{t-1} - \|\boldsymbol{p} - \boldsymbol{w}\|_t$, reach bonus, time penalty.
- **Obstacle navigation**: up to 24 kinematic cylinders ($r \in [0.15, 0.45]\,\mathrm{m}$) rejection-sampled with a minimum gap of $0.9\,\mathrm{m}$, goal 11–12 m ahead; progress, goal, heading-alignment (camera looks where the vehicle flies), altitude-band and proximity terms. A curriculum increases the maximum obstacle count from 8 to 24 when the running success rate exceeds 70 %. Observation variants: 32-ray range ring, analytic depth (64×48, HFOV 87°), RTX depth, RGB, RGB-D.

5.3 Domain randomization

Startup (written to PhysX): mass $\times[0.9, 1.1]$, inertia $\times[0.85, 1.15]$ per axis, CoM offset $\pm 1\,\mathrm{cm}$. Per episode (model side): $k_f \times [0.9, 1.1]$ per rotor, $k_m \times [0.85, 1.15]$, motor time constants $\times[0.7, 1.6]$, $\omega_{\max} \times [0.92, 1]$ (battery sag), drag $\times[0.5, 1.5]$, payload $-5 \ldots + 10\,\%$ of weight, sensor-noise scale $\times[0.5, 1.5]$, action delay 0–2 and estimator delay 0–2 physics steps. The *v1* tasks (`Quadsim-{Hover,Track,Waypoint}-v1`) add the corrections found in the PX4 SITL cross-check (Section 7): the EKF-like estimate model, action and estimator delays of 0–4 steps (0 ms to 20 ms), noise scale $\times[0.5, 2]$, a per-episode rate-PID gain scale $\times[0.6, 1.4]$ and a thrust-map gain around hover, $C = C_h + g\,(C_{\mathrm{cmd}} - C_h)$ with $g \in [0.8, 1.7]$, which represents a thrust-curve or calibration error of the deployed thrust mapping. `Quadsim-Hover-v2` additionally restores the v0 white-noise level and penalizes the change of the collective command (Section 7.5).

5.4 Learning algorithm

PPO [8] (rsl_rl 3.1.2) with GAE ($\gamma = 0.99, \lambda = 0.95$), adaptive learning rate (KL target 0.01), 32 steps per environment, empirical observation normalization. State policies: actor MLP 256–256–128, critic 512–256–128 (ELU). Vision policies: a 3-layer CNN encoder (32/64/64 channels) to a 128-d feature, fused with the normalized state in a 256–128 actor; the critic is a state-only MLP on privileged inputs.

6 Validation and experiments

Table 1 summarizes the acceptance status of every milestone; the subsections give the evidence.

6.1 M1: simulator foundation

Sixty-four rigid bodies in free fall for 0.5 s at $\Delta t = 2.5\,\mathrm{ms}$ end at $|z - z_{\mathrm{symplectic}}| = 9.5 \times 10^{-7}\,\mathrm{m}$ (float32 resolution) and $|z - z_{\mathrm{analytic}}| = 6.1\,\mathrm{mm}$, which is exactly the expected first-order bias $g\,t\,\Delta t/2$; this identifies PhysX's integrator as symplectic Euler, which the reference simulator reproduces. The same test passes inside the Docker container.

Milestone	Acceptance criterion (executed)	Status
M1 Isaac Sim/Lab	headless GPU boot; PhysX free fall vs symplectic Euler $< 1\,\text{mm}$ (got $9.5 \times 10^{-7}\,\text{m}$)	**pass**
M2 Dynamics	unit tests (all modules); PhysX vs reference $< 1\,\text{cm}/1.5\,\text{s}$; inertia exact; motor/yaw models $< 3\,\%$	**pass**
M3 Sensors, recording, visualization	sensor statistics; rendered vs analytic depth (median $< 2\,\text{cm}$); HDF5 round trip; GUI and video	**pass**
M4 State-based RL	training completes with checkpoints; evaluated vs classical baselines	**pass**
M5 Vision RL	depth policy trained and evaluated (incl. rendered depth); RGB/RGB-D pipelines execute	**pass***
M6 DR & Sim-to-Real interfaces	DR ablation; ONNX = PyTorch to 10^{-5}; deployment-in-the-loop; ROS 2/PX4 bridge	**pass**†
M7 Docker, benchmark, docs	container runs tests + GPU smoke test; benchmark table; README; report	**pass**
PX4 SITL	policies fly PX4 v1.16 SITL as offboard controllers; quadsim-vs-SITL comparison of hover, steps, tracking, waypoints, navigation; bridge safety gating verified	**pass**‡

Table 1: Milestone acceptance summary. *Criteria met; the depth policy's success (61–67 %) is moderate and no RGB(-D) policy performance is claimed. †DR showed no significant success gain on hover; the bridge was validated in PX4 SITL afterwards. ‡Hover, tracking and waypoint transfer at the control level; the velocity navigation policy needs a matched velocity loop (Section 7).

6.2 M2: quadrotor dynamics

Unit tests (CPU, no simulator) verify quaternion algebra, FLU sign conventions of roll/pitch/yaw torques, the exact motor step response (63.2 % after τ, separate rise/fall constants, discretization invariance), rotor and body drag, ground effect and gyroscopic torque, analytic free fall, constant-torque yaw, conservation of angular momentum and energy for a tumbling body (Dzhanibekov effect), first-order convergence of the semi-implicit integrator to RK4, closed-loop position steps with the full cascade ($< 5\,\text{cm}$), recovery of the Crazyflie from 46° tilt and 3.7 rad/s, SE(3) circle tracking ($< 10\,\text{cm}$ RMSE), and allocation/desaturation properties. Table 2 lists the Isaac Sim integration tests; Fig. 2 shows the open-loop comparison.

Test	X500	Crazyflie	Criterion
PhysX mass / inertia rel. error	0e+00 / 0e+00	0e+00 / 0e+00	$< 10^{-4}$
Hover altitude drift over 2 s [mm]	0.00	0.00	< 5
Thrust step: accel. RMS err / peak	0.019 %	0.026 %	$< 2\,\%$
Yaw accel. vs τ_z/J_z	0.0036 %	0.0077 %	$< 3\,\%$
Open loop 1.5 s, pos. err vs semi-implicit ref. [mm]	0.40	2.07	< 10
Open loop, att. err vs semi-implicit ref. [deg]	0.006	0.066	< 0.5
Open loop, pos. err vs RK4 ref. [mm] (PhysX discretization)	3.3	8.9	info
Closed-loop velocity cascade, PhysX vs ref. [mm]	12.3	9.2	< 30

Table 2: PhysX integration of the quadrotor model vs analytic models and the independent reference integrator (`scripts/validation/m2_dynamics_validation.py`). Sensor noise, latency and randomization disabled.

6.3 M3: sensors, recording and visualization

Statistical unit tests confirm the IMU white-noise standard deviation at 200 Hz ($\sigma_c\sqrt{f}$, within 5 %), turn-on bias, exact latency, zero specific force in free fall and the correct gravity direction when tilted, ISA barometric conversion (100 m $\rightarrow$ 100 129 Pa), Gauss–Markov stationary statistics, depth-noise statistics, exact ray–primitive intersections and agreement of the Warp ray-casting kernel with the

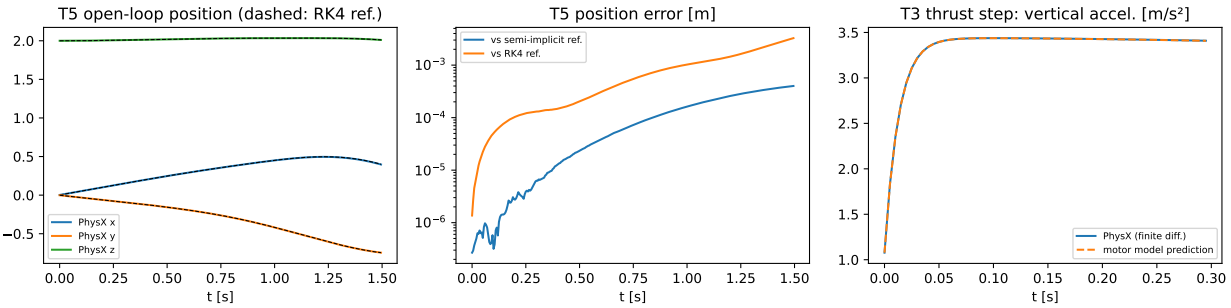


Figure 2: X500: open-loop multisine rotor excitation (left), position error of PhysX vs the reference integrator (middle; semi-implicit reference = same discretization, RK4 = integration error of PhysX at 200 Hz) and thrust-step response (right).

PyTorch implementation. A recording plot revealed a bug (no-return depth pixels received $\sigma \propto z^2$ noise); it was fixed and a regression test added.

Rendered vs analytic depth. For 64 random poses among 24 obstacles (96×72, HFOV 87°, noise off), RTX-rendered and analytic depth agree as in Table 3. Residuals concentrate on silhouette pixels (anti-aliasing) and cylinder tessellation. The comparison also exposed that the initial camera mount saw the X500's front rotor discs (3 % of the image); the default mount was moved to $(0.15, 0, -0.06)$ m, which has no self-occlusion at zero pitch.

Camera pitch	median [mm]	p90 [cm]	hit/miss agree	self-occl.	pixels
0°	7.2	2.4	99.6 %	0.0 %	337361
15°	9.0	2.3	99.5 %	5.6 %	254392

Table 3: RTX TiledCamera depth vs analytic ray-cast depth (`scripts/validation/m3_depth_consistency.py`). Self-occluded pixels (vehicle rotors, rendered < 0.4 m, analytic > 0.45 m) are excluded from the error metrics and reported separately.

Recording. The synchronized HDF5 recorder was verified by round trips for a state task (tracking, SE(3) baseline) and a rendered-depth task (navigation, potential-field baseline): dataset shapes, monotonic timestamps with constant Δt, exact equality of the last sample with the live simulator, and the embedded configuration. **Visualization.** Headless MP4 recording and the interactive GUI viewer (chase camera, target and reference markers) work after patching an upstream Isaac Lab issue (Section 6.3.1).

6.3.1 Upstream issue found and patched

With any rendered viewport (GUI, `-video`), every RigidObject failed to initialize with "Failed to get rigid body transforms from backend" — reproduced with a plain Isaac Lab cuboid. Cause: `WrenchComposer.__init__` uses `hasattr(asset.data, "body_link_pos_w")`; `hasattr` evaluates the property, which queries PhysX before its GPU buffers exist during the PLAY callback, and the raised `Exception` is not an `AttributeError`. quadsim applies a version-gated patch that checks the attribute on the class (no PhysX query), leaving Isaac Lab's sources untouched. Isaac Lab's viewport chase camera has a similar start-up read, which the viewer avoids by enabling asset following only after the first reset.

6.4 M4: state-based reinforcement learning

All policies use CTBR actions at 50 Hz, the full randomization of Section 5.3, sensor noise and latency. Evaluation uses the deterministic policy and identical seeds (hence identical initial states and randomization) for all controllers; ± denotes a 95 % confidence interval of the mean.

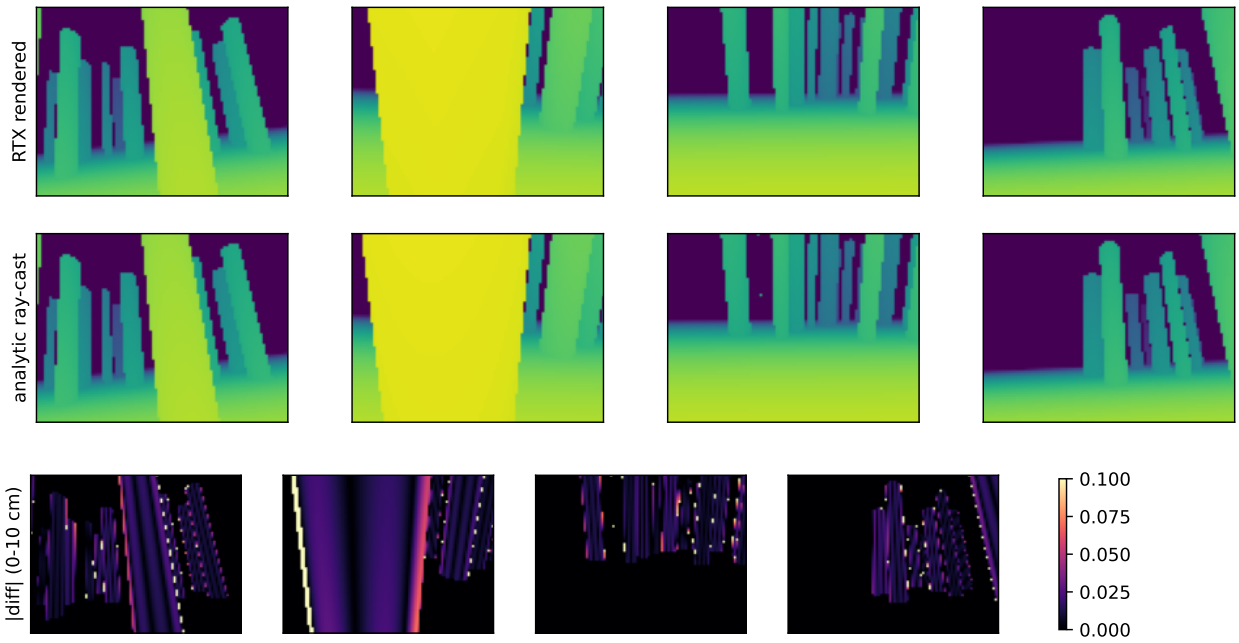


Figure 3: Rendered depth (top), analytic ray-cast depth (middle) and absolute difference clipped at 10 cm (bottom).

Hover and recovery. Training: 4096 environments, 600 PPO iterations (79 M transitions, ≈10 min on the RTX 5060). Success requires $\|e_p\| < 10$ cm and $\|v\| < 0.2$ m/s at the end of the 8 s episode. The policy also generalizes to wind (0–3 m/s mean, turbulence, gusts) and push disturbances that were *not* present during training (Table 4). The policy's yaw accuracy (4.4°) is worse than the baseline's because of the weak yaw reward weight.

Condition	Controller	success [%]	crash [%]	final pos. err [cm]	final yaw err [deg]	episodes
nominal	PPO (ours)	99.9 ± 0.2	0.0 ± 0.0	1.26 ± 0.04	4.4 ± 0.2	1000
nominal	Cascade PID	90.3 ± 1.8	0.6 ± 0.5	6.28 ± 0.95	1.5 ± 0.3	1000
wind + pushes	PPO (ours)	97.2 ± 1.0	0.0 ± 0.0	2.71 ± 0.11	4.6 ± 0.2	1000
wind + pushes	Cascade PID	80.8 ± 2.4	0.5 ± 0.4	7.60 ± 0.84	1.5 ± 0.3	1000

Table 4: Hover/recovery from up to 57° tilt, 3 rad/s, 1.5 m/s and 1.5 m offset (1000 episodes per row).

Trajectory tracking. Training: 4096 environments, 1500 iterations (197 M transitions). The aggressive reference set contains accelerations up to ≈15 m/s^2, partly beyond the X500's sustainable 13 m/s^2 (thrust-to-weight 1.69) and beyond the baselines' tilt limits; the moderate set (≤2 m/s, ≤0.35 Hz) is feasible for all controllers. The first SE(3) baseline, which mapped Lee's torque-level attitude law onto rate set-points at the 50 Hz policy rate, was unstable (53 % crashes, row †); the redesigned baseline uses Lee's position law with the onboard attitude/rate loops.

Waypoint navigation. Training: 4096 environments, 1500 iterations. Five random waypoints (separation ≥ 1.5 m, 0.3 m acceptance radius) must be passed in order within 15 s. The cascade baseline flies to each waypoint with its velocity limits (5 m/s horizontal), facing the waypoint.

6.5 M5: obstacle-avoidance navigation (range and depth)

The navigation task is the hardest: up to 24 cylinders in an 8 × 8 m field with 0.9 m minimum surface gap for a vehicle of 0.35 m collision radius, velocity-command actions and a forward-facing camera (87° HFOV) for the depth policy. Evaluation uses the full density (curriculum off, 4–24 active obstacles, mean ≈12).

Reference set	Controller	RMSE [cm]	max err [cm]	crash [%]	episodes
aggressive (≤4 m/s)	PPO (ours)	12.6 ± 1.5	40.6 ± 3.0	0.2 ± 0.4	500
aggressive (≤4 m/s)	Cascade PID	46.6 ± 3.7	104.0 ± 7.3	3.0 ± 1.5	500
aggressive (≤4 m/s)	SE(3) geometric	49.0 ± 3.8	97.6 ± 6.8	3.2 ± 1.5	500
aggressive (≤4 m/s)	SE(3) v1 (torque-level @50 Hz)†	114.2 ± 5.0	221.8 ± 9.0	53.2 ± 4.4	500
moderate (≤2 m/s)	PPO (ours)	5.8 ± 0.2	29.0 ± 0.8	0.0 ± 0.0	500
moderate (≤2 m/s)	Cascade PID	17.6 ± 1.4	50.6 ± 3.8	0.2 ± 0.4	500
moderate (≤2 m/s)	SE(3) geometric	22.7 ± 1.4	52.8 ± 3.8	0.2 ± 0.4	500

Table 5: Trajectory tracking (500 episodes of 10 s per row). Maximum errors include the initial transient (start offset up to 0.3 m).

Condition	Controller	success [%]	waypoints [%]	time [s]	crash [%]	episodes
nominal	PPO (ours)	99.4 ± 0.7	99.7 ± 0.3	6.66 ± 0.11	0.6 ± 0.7	500
nominal	Cascade PID	91.6 ± 2.4	97.0 ± 1.1	10.38 ± 0.15	2.0 ± 1.2	500

Table 6: Waypoint navigation (500 episodes). Time is measured over successful episodes only.

Reward iteration (documented). The first reward (proximity weight 0.2, collision −10, $\gamma = 0.99$) produced a policy that flies at the 3 m/s limit and accepts collisions (52.9 % success, 46.3 % collision): the proximity term contributed ≈ -0.005 per episode and the 2 s effective horizon favoured speed. Reward v2 (proximity 2.0, collision −20, $\gamma = 0.995$, everything else unchanged) was used for both the range and the depth policies. The classical potential-field baseline (same range ring) rarely collides but gets trapped in local minima.

Policy input / eval	Controller	success [%]	collision [%]	time [s]	min clear. [cm]	episodes
range ring (paired run)	Potential field	39.4 ± 4.3	4.0 ± 1.7	10.50 ± 0.49	70 ± 2	500
range ring, reward v1	PPO (ours)	57.8 ± 4.3	41.2 ± 4.3	3.71 ± 0.06	45 ± 2	500
range ring, reward v2	PPO (ours)	84.4 ± 3.2	15.2 ± 3.2	4.26 ± 0.06	53 ± 2	500
depth 64x48 (analytic, train dist.)	PPO (ours)	63.8 ± 4.2	33.8 ± 4.2	4.29 ± 0.06	50 ± 2	500
depth, zero-shot on RTX-rendered	PPO (ours)	60.0 ± 5.6	37.7 ± 5.5	4.31 ± 0.09	49 ± 3	300
range ring (independent run)	Potential field	36.6 ± 4.2	4.2 ± 1.8	10.07 ± 0.44	70 ± 2	500

Table 7: Obstacle navigation at full density (curriculum off). The depth policy (trained 900 iterations on analytic depth) is evaluated both on its training sensor and zero-shot on RTX-rendered depth. Minimum clearance is measured from the vehicle centre to the nearest cylinder surface (collision below 35 cm).

Findings (Table 7):

- **Reward v2** raised the range-ring policy from 57.8 to **84.4 %** success and cut collisions from 42.2 to 15.6 % at nearly the same speed (4.3 s to the goal).
- **Classical baseline.** The potential field rarely collides (7 %) but is trapped in local minima in 53–56 % of the episodes and is 2.5× slower (36.6–39.4 % success in two runs). An earlier version of this report gave 49.8 and 39.3 % for the two runs and attributed the spread to unpaired episodes; the actual cause was the evaluator's sampling bias (Section 8.4), which affected the run with two episodes per environment.
- **Depth policy.** With the forward-looking camera only (87° HFOV, no memory), the CNN policy reaches 63.8 % success on its training sensor, below the 360° range policy (84.4 %). A plausible but *unverified* explanation is that a memoryless policy cannot account for obstacles that have left the camera's field of view; a recurrent policy or a wider camera are the natural next experiments.
- **Analytic vs rendered depth.** Evaluated zero-shot with RTX-rendered depth, the depth policy achieves 60.0 % success / 39.7 % collisions against 63.8 / 35.8 % on the analytic depth it was

trained on, a difference within the confidence intervals (no significant degradation). Together with the pixel-level agreement of Table 3, this justifies training depth policies with the fast analytic sensor (83 k vs 6.8 k env-steps/s, Table 9).

- **RGB / RGB-D.** The rendered RGB and RGB-D environments, image augmentation and the CNN actor-critic execute end to end (10-iteration PPO run with 64 environments). No RGB(-D) policy performance is claimed.

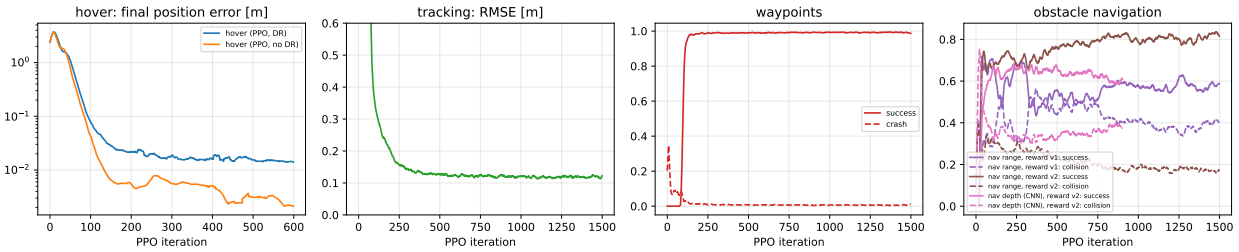


Figure 4: Training curves (episode-end metrics of the stochastic training policy, smoothed). The two hover curves are each measured on their own training distribution (the no-DR policy trains without noise) and are not directly comparable; see Table 8 for the controlled comparison. Navigation success is measured at the current curriculum density: the maximum obstacle count rises from 8 to 24, reached at iteration 112 (range v1), 124 (range v2) and 501 (depth v2); the early decline of the depth curve reflects the densifying forest. The depth run was stopped at iteration 900 (400 iterations at full density, success flat over the last 300) for compute-time reasons.

6.6 Domain-randomization ablation

An identical hover policy was trained without any randomization, sensor noise or latency (*w/o DR*) and compared with the default policy (*with DR*) under three evaluation conditions: *ideal* (the no-DR training distribution), *randomized* (the DR training distribution) and *harsh* (outside both training distributions: mass $\times[0.8, 1.25]$, motor time constants $\times[1.5, 2.5]$, thrust coefficients $\times[0.8, 1.0]$, sensor noise $\times[1.5, 2]$, 10–20 ms action delay, wind and gusts). The DR policy has lower position error in

Eval. condition	Policy	success [%]	crash [%]	final pos. err [cm]	final yaw err [deg]	episodes
ideal	PPO with DR	100.0 ± 0.0	0.0 ± 0.0	0.22 ± 0.00	3.7 ± 0.1	1000
ideal	PPO w/o DR	100.0 ± 0.0	0.0 ± 0.0	0.28 ± 0.00	9.8 ± 0.2	1000
randomized	PPO with DR	99.9 ± 0.2	0.0 ± 0.0	1.26 ± 0.04	4.4 ± 0.2	1000
randomized	PPO w/o DR	99.9 ± 0.2	0.0 ± 0.0	1.92 ± 0.07	10.2 ± 0.2	1000
harsh	PPO with DR	78.7 ± 2.5	4.0 ± 1.2	15.93 ± 2.32	19.7 ± 2.3	1000
harsh	PPO w/o DR	78.2 ± 2.6	3.4 ± 1.1	16.78 ± 2.26	21.7 ± 1.9	1000

Table 8: DR ablation on the hover task (1000 episodes per row, identical seeds).

every condition (e.g. 1.26 vs 1.92 cm under the training randomization, 15.9 vs 16.8 cm under harsh conditions), but crash rates (4.0 vs 3.4 %) and success rates (78.7 vs 78.2 %) under harsh conditions are within the 95 % confidence intervals. We therefore do *not* claim a significant robustness gain from DR for this task. A plausible explanation is the CTBR interface: the onboard rate loop closes around the (real) gyro and absorbs much of the inertial and actuator variation, a robustness property reported for CTBR policies in [3]. The yaw error of the DR policy is 2.3–2.7× lower in the ideal and training conditions but similar under harsh conditions; with one seed per arm this may be a training-run effect rather than a DR effect. Multi-seed ablations, and ablations with the direct motor interface where DR is expected to matter more, are planned (Section 10).

6.7 M6: export and deployment pipeline

The exported ONNX graph (normalizer + actor + clamp) matches PyTorch to 3.6×10^{-7}. In the deployment-in-the-loop test the simulator is flown by the ONNX policy whose observation is rebuilt every step by the NumPy `ObservationBuilder` from onboard quantities only. Over 512 hover episodes the maximum observation error is 9.5×10^{-7}, the maximum action error 1.0×10^{-6}, and the success rate (99.8 %) and final error (1.29 cm) equal those of the PyTorch policy. Frame conversions to PX4 (NED/FRD) are unit-tested, including the consistency of a physical body vector through both quaternion conventions.

6.8 Off-policy baseline: SAC

To support algorithms beyond PPO, an off-policy Soft Actor-Critic trainer (skrl 1.4.3) runs on the same environments (`scripts/rl/train_sac.py`). On the hover task with 256 environments and 30 000 vectorized steps (7.7 M transitions, 30 k gradient updates, 21 min) the episode return rose monotonically from −11.2 to +1.7 (still increasing; PPO reaches ≈ 11 on the same reward). The deterministic policy never crashed but ended 1.13 m from the target on average (0 % success over 1024 episodes). SAC is therefore *executing and learning* but *not competitive* within this budget; a higher update-to-data ratio and tuning are required. This baseline is superseded by the framework's GPU-native SAC/TD3, which reach 99.9 and 99.6 % hover success (Section 8, Table 18).

6.9 M7: throughput, memory and containerization

Table 9 lists environment-stepping throughput (no learning) on the idle RTX 5060 (8 GB); VRAM is the whole process including Isaac Sim. State-based tasks scale almost linearly up to 8192 environments (0.44 M env-steps/s, 1.76 M physics steps/s at 200 Hz). Analytic depth (Warp kernel, 3072 rays per environment) costs only ≈1.6× relative to the state task at 2048 environments, whereas RTX-rendered cameras are limited to a few thousand env-steps/s. End-to-end training throughput (with PPO updates) was ≈190 k (hover, 4096 envs), 117 k (range navigation) and 16 k env-steps/s (depth CNN, 2048 envs). The Docker image (`docker/Dockerfile`, based on `nvcr.io/nvidia/isaac-lab:2.3.2`)

Task	envs	k env-steps/s	ms / step	VRAM [GB]
`Hover-v0`	256	16.1	15.92	2.27
`Hover-v0`	1024	63.0	16.26	2.96
`Hover-v0`	4096	237.0	17.29	3.72
`Hover-v0`	8192	438.9	18.66	3.86
`Nav-Depth-v0`	256	14.5	17.67	2.33
`Nav-Depth-v0`	1024	52.5	19.51	3.24
`Nav-Depth-v0`	2048	83.1	24.65	4.41
`Nav-Depth-Render-v0`	64	2.1	29.85	4.17
`Nav-Depth-Render-v0`	128	3.8	33.77	4.85
`Nav-Depth-Render-v0`	256	6.8	37.79	5.60
`Nav-RGBD-v0`	64	2.1	30.66	4.19
`Nav-RGBD-v0`	128	3.8	33.28	4.87

Table 9: Environment throughput and process VRAM (`scripts/tools/benchmark.py`).

builds, and the containerized test service passes all unit tests and the Isaac Sim GPU smoke test. Interactive GUI visualization (chase camera, target/reference markers) and headless MP4 recording of trained policies were verified after the patch of Section 6.3.1.

7 PX4 SITL validation and sim-to-sim analysis

The exported policies were flown as PX4 offboard controllers against PX4 v1.16.2 software-in-the-loop (SITL) with the Gazebo Harmonic `x500` model, whose rotor, motor and inertia parameters equal the

quadsim X500 preset. SITL replaces quadsim's rigid-body engine (PhysX → gz-physics), its onboard controllers (simulated rate PID and velocity cascade → PX4 `mc_rate_control`, `mc_pos_control` and the control allocator) and its estimator model (noise model → EKF2 fusing simulated IMU, GNSS, barometer and magnetometer), and adds the real deployment software path (ROS 2 node, uXRCE-DDS, PX4 time synchronization). Differences between the two simulators therefore expose the assumptions of the training environment that a real flight would also violate. Every experiment is scripted and re-runnable (`ros2/px4_sitl/setup_px4_sitl.sh`, `scripts/sitl/`); all reported numbers are computed by the same unit-tested functions (`quadsim.deploy.closed_loop_metrics`) for quadsim and SITL.

7.1 Setup and deployment pipeline

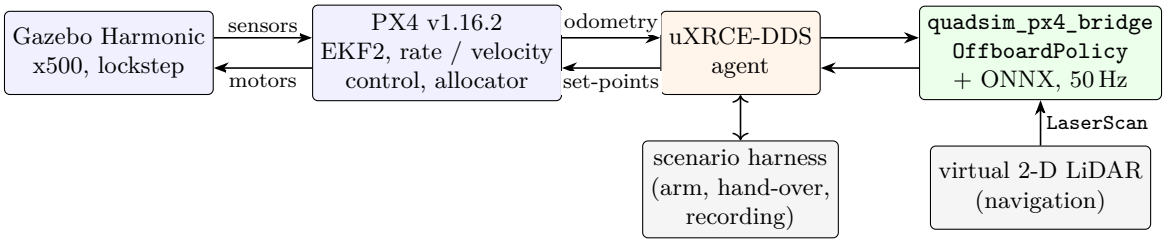


Figure 5: SITL loop. The bridge receives `VehicleOdometry` (EKF2, NED/FRD) and publishes `OffboardControlMode` plus `VehicleRatesSetpoint` (CTBR policies) or `TrajectorySetpoint` (velocity policies). The harness takes off with PX4, calibrates the thrust map, hands over to the policy, applies the task and records odometry, simulator ground truth and commands in one time base.

PX4 runs its EKF2 and controllers unmodified with airframe defaults, except `NAV_DLL_ACT`=0 (no ground station) and `EKF2_GPS_DELAY`=0 (Section 7.3). The only change to PX4 is SITL instrumentation that exports the gz ground truth over DDS for evaluation. The bridge's per-step logic is the ROS-free class `quadsim.deploy.offboard.OffboardPolicy`: it rebuilds the training observation of each task (hover goal; tracking reference $t + \{0, \dots, 1\}$ s generated onboard; waypoint sequence with the training switching rule; navigation goal plus the 32-ray range ring resampled from any `LaserScan`) and is checked against the simulator in closed loop (observation error $\leq 7.5 \times 10^{-6}$ for all four tasks, ONNX vs PyTorch success within 3 points; generalized M6 test). A scenario starts with a PX4 take-off and Hold; the goal / reference / course is defined from the current pose and the policy takes over in OFFBOARD; after the task PX4 takes back control and lands.

Thrust map. PX4 expects a normalized thrust $u \in [0, 1]$. For gz `x500` the ESC maps u to $\omega = \omega_{\min} + (\omega_{\max} - \omega_{\min})u$ and the thrust is $T = 4k_f\omega^2$, i.e. $T/T_{\max} = c(u) = (0.15 + 0.85u)^2$. The documented calibration (linear map $u = s\,C/T_{\max}$ with $s = u_h/(mg/T_{\max})$ from PX4's hover-thrust estimate u_h) is exact only at hover; there the true local gain exceeds the linear map's by

$$\frac{c'(u_h)}{c(u_h)/u_h} = \frac{2 \cdot 0.85\,u_h}{0.15 + 0.85\,u_h} = 1.61 \qquad (u_h = 0.7295). \tag{7}$$

The bridge therefore also supports an exact curve map $u = c^{-1}(s\,C/T_{\max})$ with $s = c(u_h)/(mg/T_{\max})$ (`ThrustCurve`); the calibrated $s = 1.0016$ shows that the gz model and the quadsim preset agree on hover thrust within 0.2 %.

Identification tools. Command→response dynamics are identified from closed-loop flight data by an output-error fit of a first-order-plus-dead-time model $Y(s)/U(s) = Ke^{-Ls}/(\tau s + 1)$ (grid over L and τ, least squares for K, zero-order-held commands); estimator lag by the shift minimizing $\mathrm{E}\|\hat{x}(t) - x(t - L)\|^2$; limit cycles by the RMS and Welch-spectrum peak of body rate, collective command and vertical velocity. All are unit-tested on synthetic signals with known parameters.

7.2 Defects found by SITL

- **Bridge never commanded.** PX4 1.16 versions some DDS topics (`VehicleStatus`→ `/fmu/out/vehicle_status_v1`); the original node subscribed to the unversioned name, never saw OFFBOARD and sent no set-point. Reproduced: PX4 accepted OFFBOARD for 8 s without receiving a single set-point and the vehicle drifted 3.5 m on the last attitude-controller command. Topic names are now derived from the message versions.
- **Failsafe could never trigger.** PX4 detects offboard loss only from the `OffboardControlMode` heartbeat (`COM_OF_LOSS_T`); the node published it before its stale-odometry and geofence checks. The heartbeat is now sent only while the node can command; verified: after a geofence violation PX4 left OFFBOARD after 1.03 s and, without RC, landed.
- Packaging (`setup.cfg`), a *last-action* input containing never-applied commands before the hand-over, NaN/frame checks of the odometry.
- **Measurement artefacts.** PX4's gz ground-truth velocity and "angular velocity" are backward differences (the latter of Euler angles); truth rates are recomputed by central differences of the ground-truth quaternions. rclpy's multi-threaded executor dropped ≈35 % of the messages in the recorder (PX4 itself received every set-point).

7.3 Estimator

Table 10 summarizes EKF2's error against the gz ground truth. In hover it is dominated by a per-flight bias (2–4 cm) and a slow, correlated component (std 1.4 cm, correlation time 2–3 s); the white part is ≈1 mm and the attitude error ≈0.2 mrad. quadsim's v0 odometry model is the opposite: mostly white (1 cm position, 3 cm/s velocity, 10 mrad attitude). In motion, EKF2 led the truth by 0.11 s: its default GNSS delay compensation (`EKF2_GPS_DELAY`=110 ms, correct for real receivers) assumes a latency the gz GNSS does not have. Matching it (0 ms) removed the lead and halved the truth-referenced tracking error (14→7 cm). These statistics define the EKF-like model of the v1 training distribution.

Quantity	PX4 EKF2 in SITL
Per-flight position bias $\lvert\bar{e}_p\rvert$ (x / y / z) [cm]	1.8 / 1.8 / 3.9
Slow position error (<0.5 Hz) std [cm]	1.4 / 1.4 / 1.5
Slow position error correlation time [s]	1.7 / 1.9 / 3.2
Fast position error std [mm]	1.3 / 1.5 / 1.1
Velocity error std [cm/s]	0.94 / 1.05 / 0.41
Velocity error correlation time [s]	0.81 / 0.35 / 2.14
Roll / pitch error std [mrad]	0.19 / 0.20
Circle flight, `EKF2_GPS_DELAY`=110 ms: lag x / y [ms], RMS error [cm]	-110 / -99, 12.0
Circle flight, `EKF2_GPS_DELAY`=0 ms: lag x / y [ms], RMS error [cm]	-3 / -1, 6.3

Table 10: PX4 EKF2 error vs gz ground truth (`scripts/sitl/estimator_stats.py`, `docs/results/sitl_estimator.json`).

7.4 Hover and set-point steps

The scenario of Table 11 holds the hand-over pose for 30 s (RMS over 10–30 s), then steps the goal by 1 m in x and in z; quadsim flies the same schedule with 64 environments starting at the goal.

Findings: (i) *Control performance transfers*: referenced to the estimate, SITL holds the pose to 0.8 cm to 1.0 cm RMS (quadsim 1.9 cm) and the z step rises and settles as in quadsim; the smoother EKF2 estimate produces much quieter commands than quadsim's white-noise odometry. (ii) With PX4's default `EKF2_GPS_DELAY` the x step settled slower (1.86 s, the position dipping back to ≈0.9 m); with the GNSS delay matched to gz the same policy settles in 1.01 s to 1.04 s (quadsim 1.07 s) – the dip was an estimator artefact of the SITL configuration, not a control difference. (iii) *Heave*

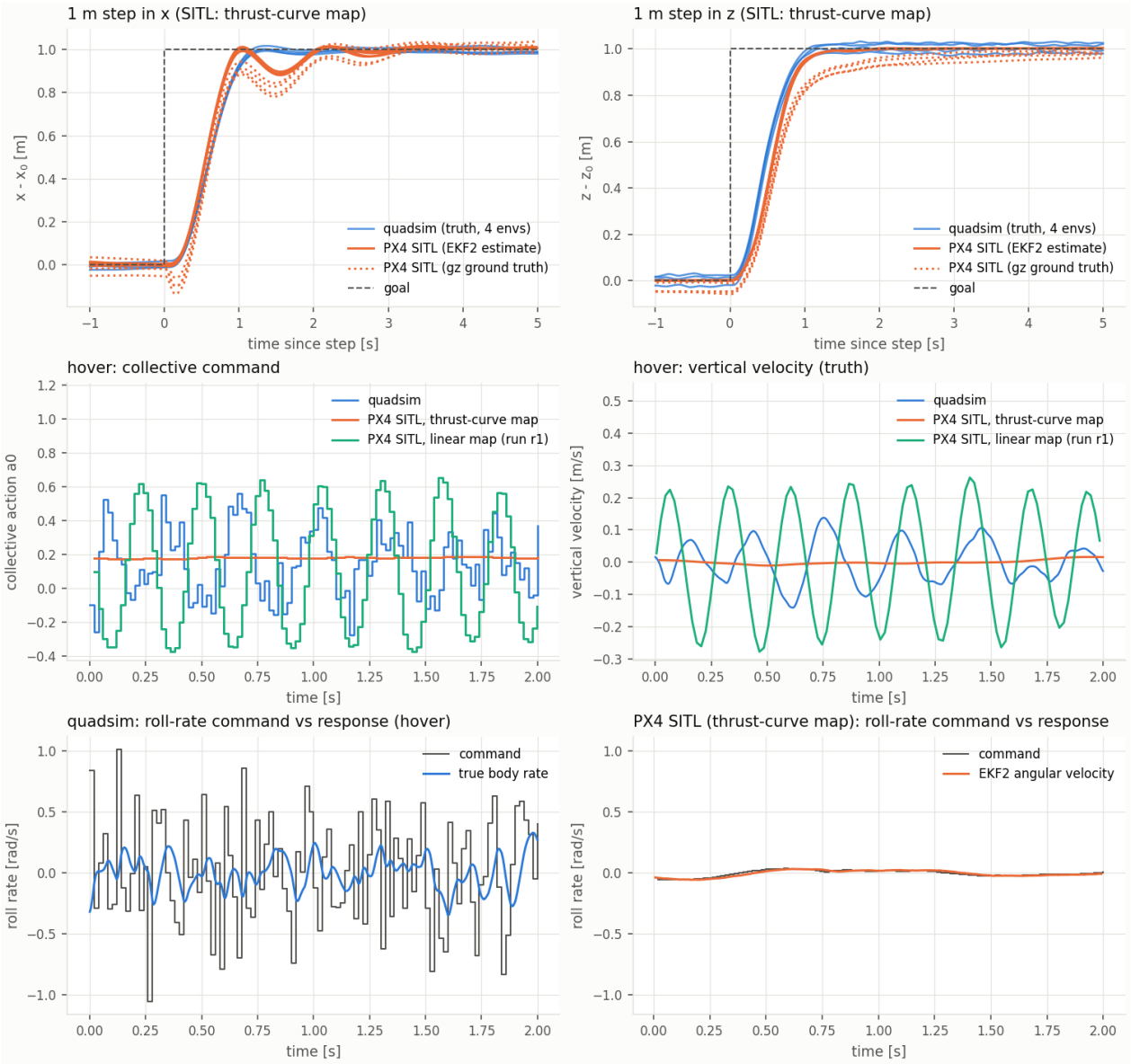


Figure 6: Top: 1 m steps (quadsim truth, SITL estimate and truth). Middle: hover collective command and vertical velocity – with the linear thrust map PX4 SITL shows a 4 Hz heave limit cycle that the thrust-curve map removes. Bottom: roll-rate command and response (quadsim commands chatter because of the white odometry noise).

Metric	quadsim nominal	SITL linear map (3)	SITL curve map (4)	SITL linear + geometry fix (2)
Hover RMS error, truth [cm]	1.59 ± 0.46	4.12 ± 1.94	6.00 ± 1.11	4.71 ± 1.06
Hover RMS error w.r.t. estimate [cm]	1.88 ± 0.02	1.02 ± 0.13	0.83 ± 0.04	1.01 ± 0.22
Estimate vs truth, hover RMS [cm]	2.28 ± 0.31	3.84 ± 2.08	5.81 ± 1.19	4.51 ± 1.08
x step w.r.t. estimate: rise 10–90 % [s]	0.65 ± 0.01	0.54 ± 0.01	0.55 ± 0.01	0.54 ± 0.00
x step w.r.t. estimate: settling (5 cm) [s]	1.07 ± 0.02	1.84 ± 0.02	1.86 ± 0.02	1.86 ± 0.02
z step w.r.t. estimate: rise 10–90 % [s]	0.63 ± 0.01	0.64 ± 0.01	0.64 ± 0.01	0.62 ± 0.02
z step w.r.t. estimate: settling (5 cm) [s]	1.00 ± 0.03	1.07 ± 0.05	1.02 ± 0.02	1.07 ± 0.03
Rate cmd→rate: delay roll [ms]	13 ± 4	23 ± 1	31 ± 1	15 ± 5
Rate cmd→rate: delay pitch [ms]	13 ± 4	13 ± 2	10 ± 1	10 ± 2
Rate cmd→rate: τ roll [ms]	53 ± 2	47 ± 2	38	38 ± 3
Rate cmd→rate: τ pitch [ms]	52 ± 0	39 ± 1	42 ± 1	41 ± 3
Hover collective a_0 std	0.174 ± 0.018	0.218 ± 0.139	0.005 ± 0.000	0.178 ± 0.174
Hover vertical velocity RMS, truth [cm/s]	5.4 ± 1.1	10.1 ± 6.2	1.0 ± 0.0	8.8 ± 7.8
Odometry age at policy step, median [ms]	–	11.9 ± 0.9	8.0 ± 1.8	12.5 ± 3.2

Table 11: Hover policy (v0, seed 42), quadsim vs PX4 SITL (number of SITL runs in brackets; ± = std over envs / runs). "w.r.t. estimate" measures what the policy regulates; truth-referenced numbers include EKF2's error. These SITL runs used PX4's default `EKF2_GPS_DELAY`=110 ms, which slows the x step (see (ii)); later runs use 0 ms. Source: `docs/results/sitl_comparison.json`.

limit cycle: with the linear thrust map, 3 of 5 runs developed a persistent 3.75 Hz to 4 Hz vertical oscillation (collective std 0.30–0.36, vertical velocity 0.14–0.17 m/s RMS) from the hand-over on; with the curve map 0 of 4 runs did. Quadsim shows the same lightly damped mode as noise-driven heave; the 1.6× local thrust gain of the linear map turns it into a limit cycle. (iv) *Rate loop*: PX4's pitch response matches quadsim's delay (8–14 vs 13 ms) with a smaller time constant (38–44 vs 53 ms); roll is slower (22–32 ms) because the x500 airframe's allocation geometry (`CA_ROTOR*`=0.13/0.20–0.22 m) differs from the gz rotor positions (±0.174 m); setting the true geometry reduced the roll delay by 4–12 ms under matched conditions (2 runs). Fitted delays depend on the excitation by ±10 ms (closed-loop identification); with PX4's backward-differenced ground-truth rates all SITL delays appear ≈10 ms longer. (v) PX4's hover-thrust estimate is 0.7295 (default `MPC_THR_HOVER` 0.60); without calibration ($s = 1$, −19 % thrust) the policy still flies but holds 1.3 cm low. (vi) Odometry is 5–16 ms old at the policy step (median per run, max 23 ms) – at and above the trained 0–10 ms.

7.5 Training distributions v1 / v2 and multi-seed comparison

The findings above define the v1 distribution (Section 5.3). Hover policies were trained with three seeds on v1 and compared with three seeds of v0 (seed 42 plus two new seeds) and, after the v1 result, three seeds of a corrected distribution v2, in quadsim and in SITL with `EKF2_GPS_DELAY`=0 (Table 12). **v1 is a negative result.** Making the training estimate noise realistic (correlated, little white noise) and randomizing the thrust-map gain up to 1.7 produced policies with *more* collective chatter: 4 of 6 SITL flights with the linear map developed the heave limit cycle (v0: 1 of 6), one even with the curve map, the SITL hover error doubled (2.2 vs 1.0 cm), and success under the harsh out-of-distribution condition dropped from 78 to 67 %, while in-distribution performance was unchanged. **v2** keeps v1's correlated estimate errors, latencies and gain randomization, restores v0's white-noise level and adds a collective-rate penalty $-w\,(a_{0,k} - a_{0,k-1})^2 \Delta t$ ($w = 1$). It halves the collective chatter in quadsim (std 0.08 vs 0.15), keeps v0's hover precision, and *none* of its 9 SITL flights (3 seeds, linear and curve maps) showed the limit cycle. The price is a less damped vertical step (overshoot 7–20 % for two seeds vs 1–2 % for v0), lower harsh-condition success (70 vs 78 % for v0) and a larger yaw error. No training distribution dominates: v0 with the thrust-curve map is the best-validated deployment configuration, v2 the more robust choice when only a linear thrust map is available. Across all nine policies, the collective chatter in the quadsim thrust-gain scenario ranks the policies as SITL does, which makes it a cheap pre-deployment screen.

7.6 Trajectory tracking and waypoints

Table 13 compares the tracking policy on a circle ($r = 1$ m, 0.2 Hz, 1.26 m/s) and a figure-8 ($r = 1.5$ m, 0.15 Hz) and the waypoint policy on a fixed 5-waypoint course, all with the thrust-curve

Metric	v0 (3 seeds)	v1 (3 seeds)	v2 (3 seeds)
quadsim, v0 distribution: success [%]	99.9 ± 0.0	99.7 ± 0.4	99.0 ± 0.6
quadsim, v0 distribution: final pos. error [cm]	1.28 ± 0.07	1.83 ± 0.30	1.58 ± 0.12
quadsim, v1 distribution: success [%]	73.9 ± 2.0	74.4 ± 3.1	69.3 ± 5.7
quadsim, v1 distribution: final pos. error [cm]	6.58 ± 0.12	6.77 ± 0.23	6.84 ± 0.14
quadsim, harsh OOD: success [%]	77.8 ± 1.2	66.0 ± 4.2	70.2 ± 2.0
quadsim, harsh OOD: crash [%]	3.7 ± 0.2	3.6 ± 0.4	3.4 ± 0.4
scenario $g = 1.0$: hover RMS w.r.t. estimate [cm]	1.87 ± 0.00	2.46 ± 0.32	1.95 ± 0.05
scenario $g = 1.0$: collective std	0.149 ± 0.016	0.211 ± 0.049	0.079 ± 0.003
scenario $g = 1.6$: collective std	0.209 ± 0.026	0.286 ± 0.079	0.112 ± 0.007
scenario $g = 1.6$: vertical velocity RMS [cm/s]	8.7 ± 1.0	9.0 ± 2.3	8.6 ± 0.8
SITL linear map: runs with limit cycle	1 / 6	4 / 6	0 / 6
SITL curve map: runs with limit cycle	–	1 / 3	0 / 3
SITL linear map: collective std	0.061 ± 0.080	0.310 ± 0.171	0.062 ± 0.016
SITL linear map: hover RMS w.r.t. estimate [cm]	1.00 ± 0.10	2.17 ± 0.27	1.26 ± 0.23
SITL linear map: x step settling [s]	1.02 ± 0.02	1.05 ± 0.01	0.99 ± 0.05
SITL curve map: collective std	–	0.164 ± 0.142	0.031 ± 0.025
SITL curve map: hover RMS w.r.t. estimate [cm]	–	2.16 ± 0.74	1.29 ± 0.21

Table 12: Hover policies trained on v0, v1 and v2, mean ± std over 3 seeds. Quadsim: 1000 episodes per seed and condition; scenario: hover + steps with nominal dynamics and a thrust-map gain g (1.6 = linear map on the gz x500). SITL: limit-cycle runs = collective std > 0.1 in hover. Source: `docs/results/hover_v1.json`.

map. Referenced to the estimate, tracking in SITL equals quadsim (circle 2.2 vs 2.5 cm, figure-8 3.4 vs 3.0 cm RMSE); the truth-referenced error is 3–4× larger because of EKF2's correlated error. The waypoint course was completed in every SITL run (policy's own estimate-based switching) but took 6.7 s instead of 5.1 s; at up to 7 m/s EKF2's error reaches 30 cm, so on truth some waypoints were passed just outside the 0.3 m radius. Removing quadsim's body drag did not change its time (5.13 s); in SITL neither the true allocation geometry (6.29 s) nor roll/pitch air mode (`MC_AIRMODE`=1, 6.45 s / 6.20 s) changed it beyond the run-to-run spread. The cause remains open (candidates: EKF2's error at 7 m/s acting on the estimate-based switching and approach geometry, motor saturation at full thrust).

Task	Metric	quadsim	PX4 SITL
Tracking, circle (r 1 m, 0.2 Hz)	RMSE truth [cm]	1.8 ± 0.4	6.3 ± 0.7
	RMSE w.r.t. estimate [cm]	2.5 ± 0.1	2.2 ± 0.2
Tracking, figure-8 (r 1.5 m, 0.15 Hz)	RMSE truth [cm]	2.7 ± 0.2	9.2 ± 0.4
	RMSE w.r.t. estimate [cm]	3.0 ± 0.1	3.4 ± 0.4
Waypoints (5)	course completed	1.00	1.00
	time to finish [s]	5.13 ± 0.02	6.74 ± 0.27
	estimate error during course, RMS [cm]	–	15.2 ± 0.7

Table 13: Tracking and waypoint policies (v0), quadsim (32 envs) vs PX4 SITL (2 runs each); RMSE after the first 2 s. Source: `docs/results/sitl_tasks.json`.

7.7 Obstacle navigation through the velocity interface

The range-ring navigation policy commands velocities, which PX4's position controller tracks. Obstacles are virtual in SITL: a 2-D LiDAR node ray-casts the training ring against the cylinders of a layout from the ground-truth pose (identical to the training caster, unit-tested), and collisions are evaluated from the ground truth with the training criterion. Six layouts drawn from the full-density training distribution were flown once in SITL and by 32 quadsim environments each (Table 14).

With PX4's default velocity controller only 2 of 6 SITL flights reached the goal, against 87 % quadsim success on the same layouts. Identifying both velocity loops from the flight data explains the gap: PX4's lateral velocity response has $L = 0.21\,\mathrm{s}$, $\tau = 0.52\,\mathrm{s}$ (VAF 0.78, 6 flights) against $L = 0.14\,\mathrm{s}$, $\tau = 0.22\,\mathrm{s}$ (VAF 0.91) for quadsim's simulated cascade – PX4's default horizontal velocity gain (`MPC_XY_VEL_P_ACC`=1.8) is lower than the simulated one (3.0), so avoidance manoeuvres arrive late. Configuring PX4's velocity controller like the training cascade (`MPC_XY_VEL_P_ACC`=3.0, D=0, `MPC_Z_VEL_I_ACC`=1.0, tilt limit 0.6 rad) reduced the lateral time constant to 0.25 s (quadsim 0.22 s) and doubled the SITL success to 4 of 6 layouts; the remaining failures are layout 1 (hard in quadsim as well, 38 % success) and layout 4. The residual lateral delay (≈0.21 s vs 0.14 s) comes from PX4's attitude and rate chain and is not represented in training. With one SITL flight per layout these rates are indicative only; the lesson for deployment is that velocity-interface policies must be trained with the target autopilot's velocity-loop dynamics (identified, or randomized over a range that contains them).

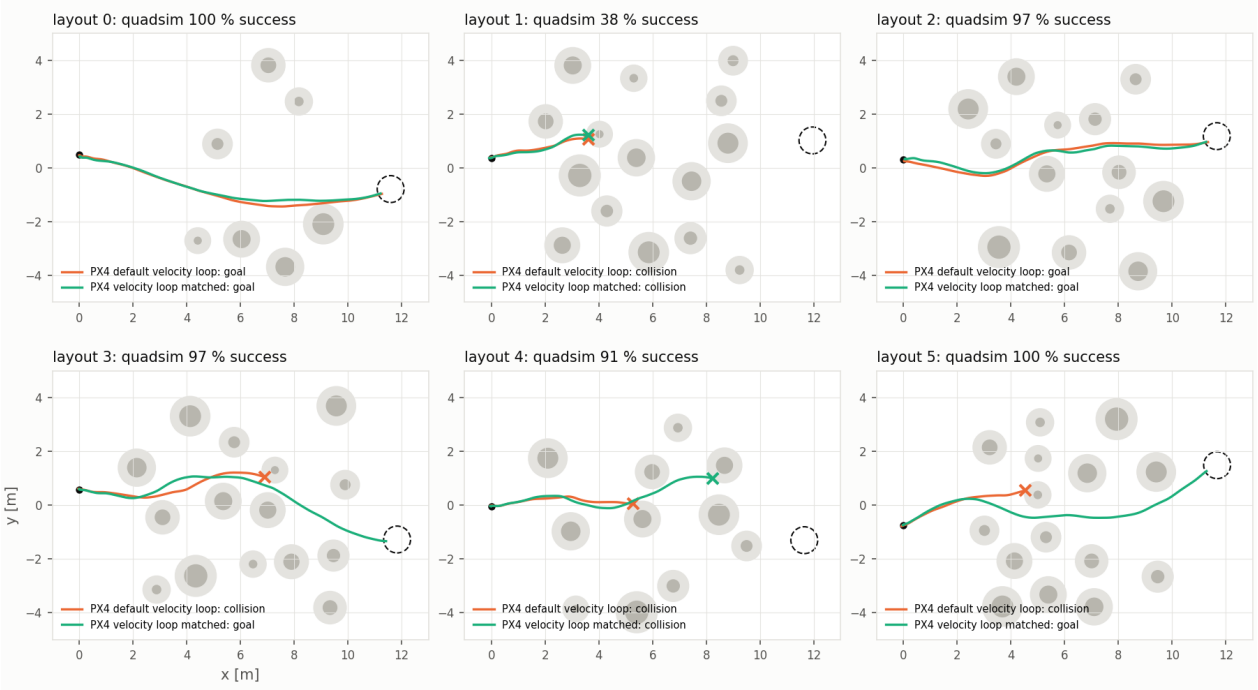


Figure 7: PX4 SITL navigation flights in layout coordinates (grey: obstacles, light grey: collision margin of the 0.35 m vehicle radius; dashed: goal region). With PX4's default velocity controller (orange) avoidance arrives late and the vehicle cuts into obstacle margins; with the velocity loop matched to the training cascade (green) 4 of 6 flights reach the goal.

Layout	obstacles	quadsim success / collision [%]	PX4 default velocity loop	PX4 velocity loop matched to quadsim
0	7	100 / 0	goal	goal
1	15	38 / 62	collision	collision
2	13	97 / 3	goal	goal
3	15	97 / 3	collision	goal
4	11	91 / 6	collision	collision
5	15	100 / 0	collision	goal

Table 14: Navigation layouts: quadsim success / collision rate (32 envs per layout) and the outcome of the PX4 SITL flight with PX4's default and with a quadsim-matched velocity controller. Source: `docs/results/sitl_tasks.json`.

7.8 Summary of the cross-check

- All CTBR policies (hover, tracking, waypoints) fly PX4 SITL as offboard controllers; at the level the policy controls (estimate-referenced) their performance matches quadsim; the bridge's safety gating is verified.
- Sim-to-sim gaps were found and quantified, each with an identified cause: EKF2's correlated error (training model too white), GNSS delay mis-configuration in SITL, the thrust-map nonlinearity (heave limit cycle; fixed by the curve map), PX4's roll/pitch-asymmetric rate loop (allocation geometry), up to 20 ms observation age, and – the largest – PX4's slower velocity loop for the velocity-interface navigation policy.
- Training on the SITL-identified distribution is not automatically better: v1 worsened the heave mode, v2 fixed it with trade-offs (multi-seed evidence, Table 12); the deployment-side fix (thrust-curve map) is the most robust.
- Not done: hardware flights, depth/RGB policies in SITL (needs a rendered camera in gz), retraining the navigation policy with a randomized velocity loop, tracking/waypoint policies on the new distributions.

8 From platform to research framework

The platform of the previous sections was built around a fixed set of tasks. An audit of that code base for reuse by other researchers found four structural limits: (i) the action interfaces were branches of one controller function, so a new interface required editing the base environment; (ii) vehicles were two hard-coded Python presets (quadrotors only for direct motor control); (iii) observations, rewards, terminations and randomizations could only be extended by subclassing a task; (iv) training was tied to one algorithm library, with a separate, non-competitive SAC script, and there were no standard benchmark conditions, no constrained or multi-agent tasks and no imitation-learning data path. This section describes the framework that replaced these limits without rebuilding the validated core, and the experiments that validate it. Architecture details are in `docs/FRAMEWORK.md`.

8.1 Extension points

Every replaceable component type has a name registry; components register themselves on import, research code registers its own (from any module or file passed as `-plugins` to every script) and selects them by name in the configuration or on the command line (Table 15). Configuration-level extensions do not need a registry: each task configuration carries *terms* with the Isaac Lab signature `func(env, **params)` (Table 16).

Table 15: Registries (`quadsim.registry`).

Registry	Selected by	Built-in components
`VEHICLES`	`dynamics.preset` (name or YAML/JSON file)	x500, crazyflie, racer5, hexa_x
`ACTION_INTERFACES`	`control.mode`	motor, ctbr, ctatt, velocity, position
`ALGORITHMS`	`-algo`	ppo (rsl_rl), sac, td3, bc, vla
`BASELINES`	`-controllers`	cascade, geometric, potential_field, swarm_potential, langnav_expert
`BENCHMARKS`	`benchmark/run.py -suite`	hover, track, waypoint, nav_range, race, swarm, langnav

Sensor terms. A term h with period P and latency D (both in policy steps) produces $o_t = h(s_{P\lfloor (t-D)/P \rfloor}) + \eta$, $\eta \sim \mathcal{N}(0, \sigma^2 I)$, with the history refilled at every episode start. The same mechanism implements a 5 Hz GNSS fix with 100 ms latency, a downward rangefinder, a 3-D LiDAR ray-casting the task's obstacle primitives, or action/gyro histories for implicit adaptation.

Table 16: Configurable MDP terms (`cfg.terms`).

Term	Semantics
observation	appended to the deployable policy observation (or the critic's, if privileged) with sampling period P, latency D, white noise, scale, clip and an optional history of H samples
reward	$w\, f(s_t)\, \Delta t$, logged per term
termination	failure, or truncation (bootstrapped)
cost	constraint cost rate $c(s_t) \geq 0$; episode cost $J_c = \sum_t c\, \Delta t$ (seconds of violation for an indicator)
event	start-up / reset / interval randomization or perturbation
curriculum	called before resets with the finished episodes' metrics (e.g. raise the initial tilt when success $> 80\,\%$)

Vehicles. A vehicle file may inherit from a preset, generate a symmetric n-rotor frame (4-rotor X frames use the PX4 Quad-X numbering), derive k_f from the thrust-to-weight ratio and estimate the inertia from a box body and point-mass motors. Vehicles are rejected if the inertia violates the triangle inequality, the thrust-to-weight ratio is ≤ 1.05 or the allocation matrix $A \in \mathbb{R}^{4\times n}$ (collective, roll, pitch, yaw) has rank < 4. The USD body, PhysX mass properties, motor and drag model, pseudo-inverse mixer with PX4-style desaturation, hover action and the motor-level action dimension follow automatically. An Isaac Lab `MultirotorCfg` (allocation matrix with rows $[F_x, F_y, F_z, M_x, M_y, M_z]$, thrust $c\,\mathrm{rps}^2$) converts with $y_i = A_{3i}$, $x_i = -A_{4i}$, $s_i = -\mathrm{sign} A_{5i}$, $k_f = c/(2\pi)^2$.

Constrained RL. With cost terms and `safety.lagrangian`, the environment optimizes the Lagrangian of $\max_\pi J_r(\pi)$ s.t. $J_c(\pi) \leq d$ by returning $\tilde{r}_t = (r_t - \lambda c_t)/(1+\lambda)$ to *any* RL algorithm and updating λ with the PID rule of Stooke et al. [14]: $e_k = \bar{J}_c - d$, $I_k = \max(0, I_{k-1} + K_i e_k)$, $\lambda_k = \min\big(\lambda_{\max}, \max(0, K_p e_k + I_k + K_d \max(0, \bar{J}_{c,k} - \bar{J}_{c,k-1}))\big)$, where $\bar{J}_c$ is the mean cost of the episodes finished since the last update (every rollout). This is the reward-penalty form (no separate cost critic).

Multi-agent RL. M consecutive environment rows form a team sharing a local frame, an episode clock and resets; each agent is one row. Any single-agent algorithm therefore trains a parameter-shared decentralized policy $\pi(a^i \mid o^i)$, while the critic receives the centralized state (true relative states and goals of all team members), i.e. a MAPPO-style [15] centralized-training, decentralized-execution scheme. A crash terminates the crashed agent and truncates its teammates; inter-agent collisions are analytic.

Off-policy learning. The original skrl SAC [9] reached 0 % hover success. Two likely causes were addressed together (not ablated separately): the tasks' rewards are Δt-scaled (about 0.02 per step), so the entropy bonus dominated, and the critic only saw the policy observation; the update-to-data ratio was also low (1/256). The framework's SAC/TD3 use asymmetric twin critics with LayerNorm on the privileged observation, running observation normalization, rewards in per-second units ($r/\Delta t$), 1024 parallel environments, batches of 16 384 and 4 gradient updates per environment step, following the many-environment recipe of FastTD3 [11]; TD3 [10] shares the implementation. Time-outs bootstrap from the current state (as in rsl_rl's PPO).

Imitation learning and VLA. Demonstrations are episode-structured HDF5 files (policy and critic observations, camera frames, actions, rewards, ground truth, estimate, instruction text, success, metrics, configuration, observation layout, tokenizer). Collection gives every environment the same episode quota, because stopping at a global count discards the episodes still running, which are the long ones. Behaviour cloning minimizes $\mathbb{E}\|\pi_\theta(o) - a^\star\|^2$; instruction tokens are embedded with a bag-of-words encoder that ignores padding and out-of-vocabulary words, and images are encoded by a CNN whose last feature map is FiLM-modulated [19] by the instruction embedding. DAgger [17]

alternates rollouts of the learner (labelled by the expert) with retraining on the aggregated data. External VLA models plug in through an adapter (images, instructions, proprioception) $\mapsto a$; datasets export to the LeRobot [21] v3.0 format and are verified by reloading them with LeRobot's own loader.

Benchmarks. A suite fixes a task, a set of conditions labelled in-distribution (training distribution) or out-of-distribution (wind and pushes, harsh dynamics, sensor noise $\times 2$–3, 20 ms to 30 ms command plus 0 ms to 30 ms estimate latency, unseen layouts or phrasings), metrics and seeds. Every controller is evaluated per condition in a fresh simulator process with the same seed; the report gives each metric with its 95 % CI and the generalization gap $\mathrm{id} - \overline{\mathrm{ood}}$ of the primary metric.

Resources. The default number of environments is lowered automatically when the free GPU memory would not fit it (memory model calibrated on the RTX 5060 measurements; never raised unless requested).

8.2 Validation of the framework

Regression. After moving the action interfaces into classes, the vehicles into a registry and adding the term machinery, the reference evaluations (hover, tracking, waypoint, range navigation; PPO policies and the cascade, SE(3) and potential-field baselines) were re-run: all 59 metrics are bit-identical to the pre-refactor run, both after the refactor and with the final code.

Unit and example tests. 124 CPU unit tests at the framework milestone (141 after the review) (registries, vehicle files and frames, allocation rank, Isaac Lab conversion, hover action of every interface on every vehicle, closed-loop position control of a hexarotor and a racer on the reference integrator, sensor-term rate/latency/history, the PID multiplier, the squashed-Gaussian log-likelihood, replay buffer, hardware model, tokenizer, datasets, BC language grounding) and an example check that runs every extension example in the simulator (Table 26).

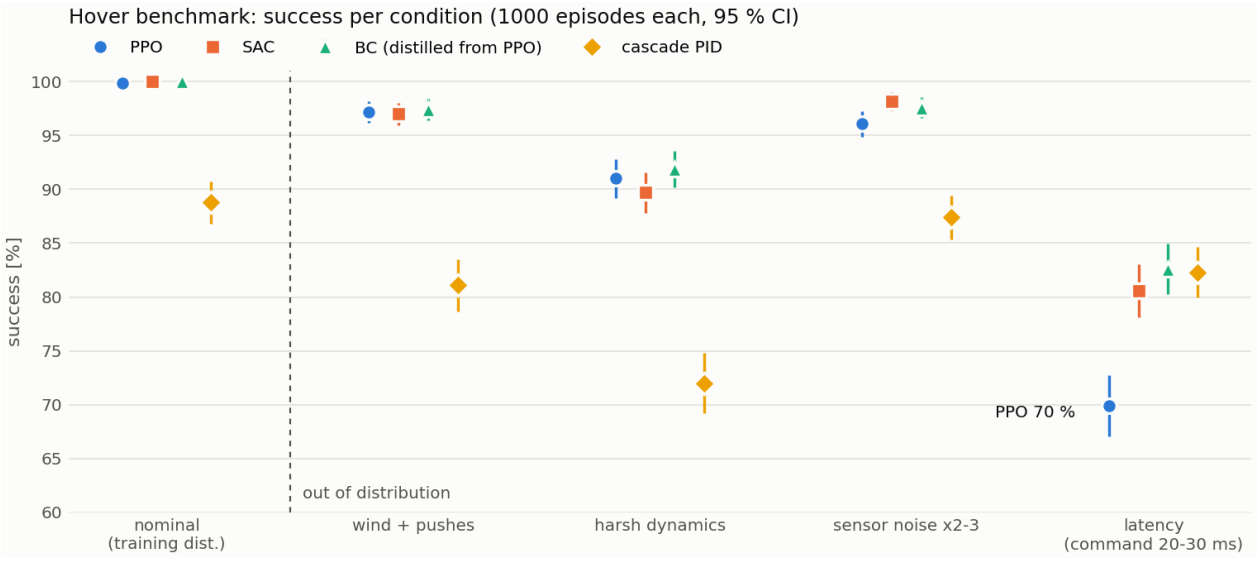


Figure 8: Hover benchmark suite: success per evaluation condition for PPO, the framework's SAC, a BC student distilled from the PPO policy and the cascade baseline (1000 episodes per point, same seed, 95 % CI; values in Table 17).

8.3 Findings

All numbers below are the values re-measured after the review (Section 8.4); racing, the swarm, the Lagrangian and the language tasks are given over three seeds.

Table 17: Hover benchmark suite: success per condition (1000 episodes each, same seed, fresh process per controller and condition) and generalization gap id−mean(ood) in points. Source: `docs/results/framework/bench_hover_algos.json`.

Controller	nominal (id)	wind_push	harsh_dynamics	sensor_noise	latency	gap
ppo	99.9 %	97.2 %	91.0 %	96.1 %	69.9 %	+11.4
sac	100.0 %	97.0 %	89.7 %	98.2 %	80.6 %	+8.6
bc_distilled	100.0 %	97.4 %	91.9 %	97.6 %	82.6 %	+7.6
cascade	88.8 %	81.1 %	72.0 %	87.4 %	82.3 %	+8.1

Table 18: Learning algorithms on hover (nominal condition, 1000 episodes). Position error in cm, yaw error in rad. Source: `offpolicy_hover.json`.

Algorithm	Success	Pos. err [cm]	Yaw err [rad]	Train [min]
PPO (rsl_rl), 4096 envs, 600 it.	99.9 %	1.26 ± 0.04	0.08 ± 0.00	–
SAC (quadsim), 1024 envs, 15k steps	100.0 %	1.67 ± 0.05	0.51 ± 0.01	21
TD3 (quadsim), 1024 envs, 15k steps	0.0 %	87.49 ± 1.77	0.63 ± 0.02	19
TD3, noise 0.05-0.15, reward scale 10, 8 upd./step	99.9 %	2.73 ± 0.24	1.06 ± 0.01	28
ARS (plugin example), 150 it.	46.8 %	11.87 ± 0.40	0.58 ± 0.04	12

Table 19: New vehicles, control interfaces and tasks defined through the extension points (1000 episodes; position error in cm). Source: `vehicles_interfaces.json`.

Configuration	Success	Pos. err [cm]	Crash
hexa_x (6 rotors, generated frame), CTBR, PPO	99.8 %	1.21 ± 0.03	0.0 %
hexa_x, cascade baseline	96.2 %	4.46 ± 0.68	0.3 %
hexa_x, motor-level (6-D action), PPO	100.0 %	1.40 ± 0.04	0.0 %
x500, acceleration interface (plugin example), PPO	99.7 %	1.40 ± 0.05	0.0 %
landing task (plugin example), PPO, first reward	0.0 %	–	0.5 %
landing task, PPO, reward v2 (time penalty, low-and-slow term)	94.1 %	–	5.9 %
landing task, cascade baseline	64.1 %	–	35.9 %

Table 20: Constrained waypoint racing: speed ≤ 2 m/s, budget 0.1 s of over-speed flight per episode (500 episodes). Source: `safe_rl.json`.

Policy	Success	Episode cost [s]	Time [s]	Crash
PPO (unconstrained)	99.4 %	5.350 ± 0.076	6.66 ± 0.11	0.6 %
cascade	91.6 %	6.503 ± 0.123	10.38 ± 0.15	2.0 %
PPO-Lagrangian (PID multiplier)	87.6 %	0.081 ± 0.017	11.89 ± 0.15	0.0 %

Table 21: Drone racing (2 laps, 7 gates, 5-inch racer); trained on random circuits. Source: `bench_race.json`.

Controller	Condition	success	lap_time	gates_passed	crash
cascade	random_tracks (id)	0.6 %	15.04 ± 0.32	6.62 ± 0.28	25.2 %
cascade	ellipse (ood)	0.0 %	15.22 ± 0.31	7.14 ± 0.26	14.4 %
cascade	figure8 (ood)	0.0 %	18.42 ± 0.18	6.06 ± 0.21	16.0 %
cascade	wind (ood)	0.6 %	15.08 ± 0.32	6.28 ± 0.29	27.8 %
ppo	random_tracks (id)	94.4 %	2.47 ± 0.02	13.49 ± 0.20	5.6 %
ppo	ellipse (ood)	99.6 %	2.49 ± 0.01	13.96 ± 0.06	0.4 %
ppo	figure8 (ood)	5.4 %	4.15 ± 0.05	3.40 ± 0.39	94.6 %
ppo	wind (ood)	95.8 %	2.47 ± 0.02	13.63 ± 0.17	4.2 %
ppo_v1	random_tracks (id)	94.8 %	2.45 ± 0.02	13.46 ± 0.21	5.2 %
ppo_v1	ellipse (ood)	98.0 %	2.49 ± 0.01	13.84 ± 0.10	2.0 %
ppo_v1	figure8 (ood)	73.8 %	3.73 ± 0.02	12.90 ± 0.50	26.2 %
ppo_v1	wind (ood)	95.4 %	2.47 ± 0.02	13.56 ± 0.19	4.6 %

Table 22: Four-agent antipodal swap (shared policy, centralized critic). Source: `bench_swarm.json`.

Controller	Condition	success	agent_collision	min_separation	time_to_goal
cascade	swap4 (id)	35.9 %	36.1 %	0.95 ± 0.02	2.28 ± 0.03
cascade	wind (ood)	29.7 %	36.9 %	0.95 ± 0.02	2.30 ± 0.03
cascade	latency (ood)	35.9 %	33.8 %	0.95 ± 0.02	2.24 ± 0.02
ppo	swap4 (id)	87.5 %	0.4 %	1.39 ± 0.02	2.04 ± 0.04
ppo	wind (ood)	89.8 %	0.8 %	1.37 ± 0.02	2.09 ± 0.06
ppo	latency (ood)	80.5 %	1.2 %	1.38 ± 0.02	2.16 ± 0.07
swarm_potential	swap4 (id)	100.0 %	0.0 %	1.28 ± 0.01	6.89 ± 0.08
swarm_potential	wind (ood)	97.7 %	0.0 %	1.29 ± 0.01	6.80 ± 0.07
swarm_potential	latency (ood)	96.1 %	0.0 %	1.24 ± 0.01	6.94 ± 0.08

Table 23: Imitation learning. Hover: RL-to-BC distillation (last column: final position error [cm]). LangNav: success with training / held-out instruction phrasings and wrong-landmark rate (training phrasing). Source: `imitation.json`.

Task	Policy	Success (train phr.)	Success (held-out)	Wrong / pos. err
hover	PPO teacher	99.9 %	–	1.26 ± 0.04
hover	BC student	99.9 %	–	1.26 ± 0.04
hover	BC student, no last action	100.0 %	–	1.32 ± 0.04
langnav (map)	scripted expert (privileged)	–	100.0 %	–
langnav (map)	BC (with last action)	15.4 %	41.0 %	0.2 %
langnav (map)	BC, instruction removed	10.8 %	10.8 %	10.0 %
langnav (map)	BC, no last action, unknown words averaged in	–	30.8 %	–
langnav (map)	same policy, unknown words ignored	81.0 %	58.2 %	0.0 %
langnav (map)	+ DAgger round 1	100.0 %	50.8 %	0.0 %
langnav (map)	+ DAgger round 2	100.0 %	56.2 %	0.0 %
langnav (map)	same data, frozen CLIP text encoder	100.0 %	97.0 %	0.0 %
langnav (RGB)	scripted expert (privileged)	100.0 %	–	0.0 %
langnav (RGB)	BC, FiLM-CNN + instruction	82.8 %	85.3 %	0.0 %
langnav (RGB)	+ DAgger round 1	100.0 %	100.0 %	0.0 %
langnav (RGB)	same data, frozen CLIP text encoder	99.7 %	99.1 %	0.0 %
langnav (RGB)	VLA adapter: colour servo	6.6 %	7.2 %	0.0 %

Table 24: Multi-seed replication: the published single-seed runs (seed 42; BC seed 0) and two more seeds with identical configurations; mean ± sample std over seeds and a stratified bootstrap 95 % CI of the mean over seeds. BC seeds vary initialization and train/validation split on the same demonstrations. Source: `multiseed.json`.

Result	Seed 42 / 0	Seed 1	Seed 2	Mean ± std	95 % CI
Race-v0, success [%], random tracks	94.4	97.2	96.8	96.1 ± 1.5	[94.4, 97.2]
Race-v0, success [%], ellipse	99.6	99.2	98.2	99.0 ± 0.7	[98.2, 99.6]
Race-v0, success [%], figure8	5.4	6.4	0.4	4.1 ± 3.2	[0.4, 6.4]
Race-v0, success [%], wind	95.8	95.8	96.6	96.1 ± 0.5	[95.8, 96.6]
Race-v1 (both directions), success [%], random tracks	94.8	96.8	95.6	95.7 ± 1.0	[94.8, 96.8]
Race-v1 (both directions), success [%], ellipse	98.0	97.2	97.2	97.5 ± 0.5	[97.2, 98.0]
Race-v1 (both directions), success [%], figure8	73.8	66.6	79.2	73.2 ± 6.3	[66.6, 79.2]
Race-v1 (both directions), success [%], wind	95.4	95.4	95.8	95.5 ± 0.2	[95.4, 95.8]
Swarm, success [%], swap4	87.5	89.1	92.2	89.6 ± 2.4	[87.5, 92.2]
Swarm, success [%], wind	89.8	77.3	89.1	85.4 ± 7.0	[77.3, 89.8]
Swarm, success [%], latency	80.5	76.6	72.7	76.6 ± 3.9	[72.7, 80.5]
Swarm, agent collisions [%], swap4	0.4	4.3	0.4	1.7 ± 2.3	[0.4, 4.3]
PPO-Lagrangian, episode cost [s] (budget 0.1)	0.081	0.126	0.108	0.105 ± 0.023	[0.081, 0.126]
PPO-Lagrangian, success [%]	87.6	0.0	85.2	57.6 ± 49.9	[0.0, 87.6]
LangNav map, BC, unknown words averaged, success [%], train phr.	81.0	68.8	78.0	75.9 ± 6.4	[68.8, 81.0]
LangNav map, BC, unknown words averaged, success [%], held-out phr.	30.8	15.6	14.4	20.3 ± 9.1	[14.4, 30.8]
LangNav map, BC, success [%], train phr.	81.0	68.8	78.0	75.9 ± 6.4	[68.8, 81.0]
LangNav map, BC, success [%], held-out phr.	58.2	64.0	62.4	61.5 ± 3.0	[58.2, 64.0]
LangNav map, DAgger-2, success [%], train phr.	100.0	100.0	100.0	100.0 ± 0.0	[100.0, 100.0]
LangNav map, DAgger-2, success [%], held-out phr.	56.2	84.2	80.0	73.5 ± 15.1	[56.2, 84.2]
LangNav map, DAgger-2 + CLIP, success [%], train phr.	100.0	100.0	100.0	100.0 ± 0.0	[100.0, 100.0]
LangNav map, DAgger-2 + CLIP, success [%], held-out phr.	97.0	95.6	98.0	96.9 ± 1.2	[95.6, 98.0]
LangNav RGB, BC, success [%], train phr.	82.8	75.3	71.9	76.7 ± 5.6	[71.9, 82.8]
LangNav RGB, BC, success [%], held-out phr.	85.3	75.3	62.8	74.5 ± 11.3	[62.8, 85.3]
LangNav RGB, DAgger-1, success [%], train phr.	100.0	100.0	100.0	100.0 ± 0.0	[100.0, 100.0]
LangNav RGB, DAgger-1, success [%], held-out phr.	100.0	95.3	100.0	98.4 ± 2.7	[95.3, 100.0]
LangNav RGB, DAgger-1 + CLIP, success [%], train phr.	99.7	100.0	99.7	99.8 ± 0.2	[99.7, 100.0]
LangNav RGB, DAgger-1 + CLIP, success [%], held-out phr.	99.1	99.7	100.0	99.6 ± 0.5	[99.1, 100.0]

Table 25: Comparisons over seeds (Agarwal et al. [12]): probability that a random seed of X beats a random seed of Y on a random condition, and the interquartile mean of the success rate [%], each with a stratified bootstrap 95 % CI. Source: `multiseed.json` (`comparisons`).

Comparison X vs Y	P(X > Y)	IQM X	IQM Y
Race v1 vs v0, figure-8	1.00 [1.00, 1.00]	73.2 [66.6, 79.2]	4.1 [0.4, 6.4]
Race v1 vs v0, all 4 conditions	0.38 [0.25, 0.53]	95.6 [95.2, 96.2]	96.1 [95.2, 96.7]
LangNav map, CLIP vs bag of words (held-out phr.)	1.00 [1.00, 1.00]	96.9 [95.6, 98.0]	73.5 [56.2, 84.2]
LangNav map, unknown words ignored vs averaged	1.00 [1.00, 1.00]	61.5 [58.2, 64.0]	20.3 [14.4, 30.8]
LangNav RGB, DAgger-1 vs BC (both phrasings)	1.00 [1.00, 1.00]	100.0 [97.7, 100.0]	76.3 [69.6, 82.2]

Table 26: Extension examples executed in the simulator (`examples_check.py`); success of the classical baseline where the check is an evaluation.

Example	Result	Baseline success
vehicle_yaml_hexa	pass	95 %
vehicle_yaml_inherit	pass	52 %
vehicle_plugin_quad250	pass	61 %
sensor_gnss	pass	–
task_landing	pass	–
task_landing_baseline	pass	58 %
reward_curriculum	pass	–
action_interface_accel	pass	–
algorithm_ars	pass	–
vla_adapter_color_servo	pass	7 %
sensor_lidar_nav	pass	43 %

- **Algorithms (Tables 17 and 18 and Fig. 8).** With the changes above SAC matches PPO's in-distribution hover success (≈100 %) in 21 instead of 8 minutes, with a slightly larger position error and a much larger yaw error (its entropy term makes it indifferent to the weakly rewarded yaw). Out of distribution, PPO is the most sensitive to latency beyond its training range (70 % vs 81–83 % for SAC and the BC student), so the benchmark exposes a property that the in-distribution number hides. TD3 with the SAC configuration did not learn; with smaller exploration noise, a smaller reward scale and 8 updates per step it reaches 99.9 %. A plug-in Augmented Random Search (linear policy) reaches 46.8 % in 12 minutes.
- **Generality (Table 19).** A six-rotor vehicle generated from a frame description trains to 99.8 % with the unchanged task, and the extension examples (vehicle files, sensor terms, a new task, a new control interface, a new algorithm) run through the unmodified scripts.
- **Racing (Tables 21 and 24).** PPO completes two laps on 96 ± 2 % of unseen random circuits at 11.9 m/s mean speed (2.5 s per lap), transfers to a fixed reference circuit and to wind, but rarely completes the figure-8 (4 ± 3 %): the training circuits only turn counter-clockwise, the figure-8 alternates directions. Retraining with half of the circuits mirrored (both turning directions, same budget) raises the figure-8 completion to 73 ± 6 % while keeping 95–98 % elsewhere — the benchmark located the gap, a one-line change of the training distribution closed most of it. The classical cascade with gate-aligned waypoints passes about 7 of 14 gates in the episode (15 s per lap, 25 % crashes); its first version (straight to the gate centre) crashed into the frames in 99 % of the episodes.
- **Multi-agent (Table 22).** The shared PPO policy coordinates four agents through the antipodal swap (agent collisions 36 % for uncoordinated straight lines, 1.7 ± 2.3 % for PPO) and is 3.4 times faster than a tuned potential field, which is, however, more reliable (100 vs 89.6 ± 2.4 % group success). The potential field's gains were selected on the training scenario after its first setting failed (0 %, 53 % collisions).
- **Constraints (Tables 20 and 24).** The Lagrangian keeps the over-speed time around its budget (0.105 ± 0.023 vs 0.1 s over three seeds) while the unconstrained policy spends 5.4 s per course above the limit; the price is a slower course and, in one of three seeds, a policy that never learned to finish the course: its multiplier grew to ≈5 within 150 iterations, before the task was learned.
- **Imitation and language (Table 23).** An RL policy distils into a BC student without loss. For language-conditioned navigation, the last-action input makes BC copy its previous command and fail to stop (causal confusion [18]); removing it and two DAgger rounds reach the expert's 100 %. A bag-of-words instruction encoder generalizes to unseen phrasings only if unknown words are ignored (62 ± 3 vs 20 ± 9 % on the same checkpoints) and then reaches 74 ± 15 % after DAgger, whereas a frozen pretrained CLIP text tower, with the same data and policy, reaches 96.9 ± 1.2 %. With the RGB camera instead of the landmark map (colours grounded in pixels), the FiLM-CNN policy reaches 77 ± 6 % and, after one DAgger round, 100 % (held-out phrasings 98 ± 3 %); its failures before DAgger were almost all collisions while approaching (a wrong pillar in 1 of 320 episodes). A model-free colour-servo adapter exercises the VLA interface (7 %: correct pillar, stopping rule outside the success zone).
- **Defects found by the checks.** The example check found that observation terms were evaluated before task buffers existed (now set up after the most-derived constructor) and that non-directory checkpoints (VLA factories) were mistaken for run directories; a crashed simulator process hung for minutes holding GPU memory (scripts now terminate after an exception). Demonstration collection stopping at a global episode count would bias datasets toward short episodes (now a per-environment quota) — the same stopping rule remained in the evaluator until the review (Section 8.4). The regression check was re-run after these fixes.

8.4 Review, corrections and multi-seed replication

Procedure. After the framework milestone the code base was reviewed by three independent reviewers (one for the core, vehicles, control interfaces and export; one for the task environments, terms and baselines; one for the learning algorithms, imitation learning, evaluation and result tooling),

each without access to the others' findings, and the documentation was checked mechanically (every referenced file and task exists). Every reported defect was reproduced before it was fixed — on the CPU wherever possible, with a unit test that fails on the previous code and passes after the fix (124 → 141 tests). The fixes were developed in a separate checkout while the multi-seed training queue kept running on the unchanged code, so that all seeds of a task were trained with the same code. Table 27 lists the defects and their effect on published results.

Table 27: Defects found by the review (effect on the results published before it, and the fix).

Defect	Effect on published results	Fix
Evaluator stopped at a global episode count and kept every finished episode: the episodes still running (the long ones) were dropped	every evaluation; largest where outcomes differ in episode length	per-environment quota, everything re-measured
Cascade baseline: velocity integrator not reset between episodes	cascade on hover, tracking, waypoints, racing	reset per episode
Initial angular velocity written to PhysX in the world frame but used as a body rate by gyro, odometry and rate loop	hover (first steps of an episode)	body-frame sample, rotated for PhysX
SAC Polyak-averaged its target critic twice per gradient step	published SAC run used $\tau_{\text{eff}} = 0.00998$ (configuration said 0.005)	one update, default $\tau = 0.01$
Off-policy resume acted randomly while refilling the replay buffer	none	policy refills the buffer
Curriculum averaged the last episode of every environment (mostly easier levels) and ran away	example only	judge each level on its own episodes
Mixer used an equal collective split also for asymmetric layouts (torque at every thrust level)	none (presets symmetric)	zero-torque split
`-plugins m.py env.a=b`: argparse attached the override to the plugin list	documented commands failed	overrides recovered
Vehicle files: `1e-5` read as a string, invalid spins, `k_f` together with thrust-to-weight, CoM outside the rotor polygon accepted	none	typed parsing, validation
Hardware check counted the simulator's own memory twice	seed-42 safe-RL run used 3456 instead of 4096 environments	own memory added back
Export: no plugins or overrides, exported network not taken from the algorithm, position contract incomplete	plug-in tasks not exportable	completed
Result collector would have read the multi-seed tag as the curated race/swarm table; stale benchmark results reused after a failure	latent	fixed sources, stale files removed
Rendered cameras: first frame of every episode showed the previous scene	RGB demonstrations (≈0.7 % stale frames)	re-render after resets
Race: 6.2 % of random circuits had overlapping gates, 0.22 % started inside a gate frame	random-circuit conditions	resampled
LangNav: a dwell completed in a collision step counted as success and crash	2 of 500 episodes	exclusive
Minor: unbounded cost list without Lagrangian, multiplier integral wind-up, silent reward-term override, ignored `speed_limit` override, BC on tiny datasets	none	fixed

Evaluation sampling bias. The evaluator ran N parallel environments until the number of finished episodes reached the requested E and kept all of them. With $k = E/N \approx 2$ episodes per environment this is a biased sample: an environment whose episodes end early (a crash, an early success) contributes more of them, and the episodes still running at the stopping time — by the

inspection paradox, preferentially the long ones — are discarded. Outcomes that end episodes early are therefore over-represented. In the extreme, the counter-clockwise racing policy was reported at 0 % on the figure-8 because its few successful two-lap episodes (≈7.4 s) had not finished when the crashed environments (≈1 s) had produced the requested count. The corrected evaluator takes the first $k = \lceil E/N \rceil$ episodes of every environment, which is unbiased because the inclusion of an episode no longer depends on its length (the demonstration collector already used this rule). All 127 published evaluations (and 28 of the replication) were re-run from their recorded metadata; the median change of a success rate is 0.4 points, the largest are listed in Table 28.

Regression evidence. On the 59 reference metrics, the corrected evaluator alone leaves hover and tracking unchanged (their episodes end together at the time limit) and changes navigation and the waypoint cascade; the other fixes on top change exactly the metrics they act on (hover PPO and cascade through the reset frame, the cascade on tracking and waypoints through the integrator reset) and leave the tracking, waypoint and navigation policies and the SE(3) and potential-field baselines bit-identical.

Table 28: Published vs. corrected values (largest changes; full record in `docs/results/framework/reevaluation.json`).

Result	Published	Corrected
Racing v0, figure-8 success	0 %	5.4 %
Racing v1 (both directions), figure-8 success	54.3 %	73.8 %
Racing cascade, crashes (random circuits)	40 %	25 %
Swarm, straight lines: success / collisions	18.3 / 44.6 %	35.9 / 36.1 %
Range navigation: PPO / potential field	83.0 / 49.8 %	84.4 / 39.4 %
Landing cascade success	52.0 %	64.1 %
LangNav BC with last action, held-out phrasing	20.4 %	41.0 %
LangNav DAgger-2 (bag of words) / CLIP, held-out	59.7 / 99.8 %	56.2 / 97.0 %
Hover cascade, nominal / wind + pushes	88.7 / 78.6 %	90.3 / 80.8 %

Multi-seed replication. Racing, the multi-agent swap, PPO-Lagrangian and the language-conditioned imitation policies were re-trained with two further seeds (identical code and configuration; Table 24). Training on both turning directions raises figure-8 completion from 4 ± 3 to 73 ± 6 %; the shared swarm policy reaches 89.6 ± 2.4 % group success. The Lagrangian keeps the over-speed time around its budget (0.105 ± 0.023 s for a 0.1 s budget), but one of three seeds never learned to finish the course: its multiplier grew while the policy was still learning the task. Ignoring unknown words in the bag-of-words encoder raises held-out-phrasing success from 20 ± 9 to 62 ± 3 % on the same checkpoints, and a frozen CLIP text tower reaches 96.9 ± 1.2 % against 73.5 ± 15.1 % for the bag of words after two DAgger rounds.

8.5 Tools for VLA research and rigorous reporting

Remote policies. Vision-language-action models are usually evaluated behind a policy server in their own environment: openpi's server (used for π_0 [13]) answers observation dictionaries sent over a WebSocket with msgpack-encoded numpy arrays, and Isaac Lab-Arena and BEHAVIOR evaluate models through the same interface. quadsim implements this protocol on both sides: a client adapter that queries the server every control step (camera image as uint8 HWC or exact float, proprioception, the full deployable observation and the instruction; the first action of a returned chunk is executed) and a server for quadsim checkpoints and arbitrary `infer(obs)` functions. A behaviour-cloning policy evaluated through a separate server process reproduces the local closed-loop evaluation bit for bit (every metric of 500 LangNav episodes), and a server in the LeRobot environment (NumPy 2) exchanges arrays exactly with the simulator's client (NumPy 1.26), so models with incompatible dependencies can be benchmarked on the drone tasks unchanged.

Statistics and datasets. Multi-seed results are aggregated with the metrics recommended by Agarwal et al. [12] (interquartile mean, stratified bootstrap confidence intervals over seeds, probability of improvement; Table 25). They make one nuance explicit that the means hide: training the racing policy on both turning directions wins on the figure-8 with probability 1.00 but only 0.38 over all four conditions, because it loses 0.4–1.5 points on the others. Exported LeRobot datasets carry a Hugging Face dataset card (features, provenance, verification, limitations), and the repository provides a benchmark protocol, a citation file, a licence file, a changelog and contribution rules.

Constraint warm-up and budget curriculum. To address the PPO-Lagrangian seed that never learned the course, two variants were trained (seed 42): a 300-iteration unconstrained warm-up, and the same warm-up followed by a budget that shrinks geometrically from 5 to 0.1 s over 600 iterations. Both failed (course success 2.0 and 1.2 %, over-speed 1.26 and 1.22 s per course, against 87.6 % and 0.081 s without them): the warm-up policy races with ≈5.5 s of over-speed per course, its over-speed time barely decreases once the constraint acts, the multiplier saturates at its maximum and the task collapses. In this reward-penalty form a converged policy is not steered back to the constraint; enforcing it from the start works in two of three seeds, and a separate cost critic is the next step.

9 Limitations

The following limitations are explicit; none of them is hidden by the results above.

- **No real-world flight.** All results are in simulation, including PX4 SITL (Section 7); SITL shares the quadsim motor and drag parameters, so it does not validate the motor model. In SITL, navigation obstacles are virtual (no contact physics), the waypoint course is 25–35 % slower than in quadsim for a reason not yet isolated, depth/RGB policies have not been flown (no rendered gz camera matched to the training camera), and run counts are small (2–6 flights per configuration).
- **Model fidelity.** Motor and aerodynamic parameters come from the PX4 Gazebo X500 model and the Crazyflie literature, not from identification of a physical vehicle; body-drag coefficients are engineering estimates. Rotor aerodynamics use the standard quasi-static $T = k_f\omega^2$ model (no inflow/advance-ratio dependence, no blade-element or vortex-ring-state effects); the ground effect is applied per rotor but ceiling/wall effects are not modelled. Battery dynamics are represented only by a per-episode $\omega_{\max}$ scale.
- **PhysX discretization.** PhysX integrates at 200 Hz with a first-order scheme; relative to an RK4 reference this causes 3 mm to 9 mm position error over 1.5 s of open-loop flight (Table 2), negligible for closed-loop control but relevant for open-loop prediction studies (use the reference integrator).
- **Vision.** Depth policies are trained on analytic ray-cast depth of primitive obstacles with a parametric stereo-noise model; rendered depth agrees to 7 mm median, but real sensor artefacts (edge holes, speckle, IR interference, motion blur) are not modelled. RGB/RGB-D pipelines are implemented and execute end to end, but procedural scenes lack the texture/lighting diversity needed for transferable RGB policies.
- **Evaluation statistics.** The hover policies were trained with three seeds per training distribution (Section 7.5) and racing, the swarm, PPO-Lagrangian and the language tasks with three seeds (Table 24); the other tasks use a single seed (compute budget of an 8 GB consumer GPU), so their confidence intervals quantify evaluation noise, not seed-to-seed variability — which the replication showed to be large for some results (figure-8 racing 67–79 %, held-out phrasings after DAgger 56–84 %). Swarm confidence intervals treat the four agents of a group as independent episodes, which they are not. Classical baselines use reasonable default gains, not systematically tuned gains, so the reported margins are upper bounds of the RL advantage.
- **Framework scope.** The landing, hexarotor and off-policy results use one seed; presets other than the X500 and the Crazyflie (5-inch racer, hexarotor) are engineering approximations. In all published runs, reaching the goal or finishing the course is returned as a time-out (bootstrapped

by the learners) rather than a terminal state; the option `task_end_is_terminal` changes this for new runs. The instructions come from eight training and three held-out templates over a small vocabulary; the default encoder is a bag-of-words model, the alternative a frozen pretrained CLIP text tower (no fine-tuning). No pretrained vision-language(-action) model was run or fine-tuned (the adapter interface and the LeRobot export are validated, external models are not). The off-policy policies leave a large final yaw error on hover. Autonomous exploration and explicit adaptive-control tasks are not implemented (observation histories are available as terms). The Lagrangian is the reward-penalty form without a cost critic; its multiplier is not stored in checkpoints, and it can lock a seed into a policy that never learns the task (a constraint warm-up is provided).

- **Platform issues.** Isaac Lab v2.3.2 fails to initialize rigid bodies when a viewport is rendered (GUI, video); quadsim applies a version-gated runtime patch (Section 6.3.1). Interactive GUI and 4096-environment training do not fit simultaneously in 8 GB VRAM. The first launch with rendering enabled crashed once in the shader compiler (empty cache); later launches succeeded.

10 Roadmap toward real-world deployment

Prioritized by expected impact on a successful first real flight:

1. **PX4 SITL in the loop** – done (Section 7). Follow-ups: train velocity-interface policies with an identified or randomized autopilot velocity loop; flight-test the thrust-curve map on hardware; retrain tracking and waypoint policies on the sim-to-real distribution; a rendered gz camera matched to the training camera for depth policies; HITL with the companion computer to measure the real latency chain.
2. **System identification of the target vehicle**: mass and inertia (bifilar pendulum), k_f, k_m and motor time constants (thrust stand), rotor-drag coefficient (flight logs); update the preset and center the DR ranges on the identified values; retrain with several seeds.
3. **Latency identification** of the full pipeline (estimator, companion computer, uXRCE-DDS, PX4) and matching the delay distribution used in training; add estimator-in-the-loop training using recorded EKF2 error statistics.
4. **Hardware hover/tracking experiments** (tethered, then free flight in a motion-capture arena) with the safety layer (geofence, odometry watchdog, offboard-loss failsafe), logging with the same HDF5 schema for sim/real comparison.
5. **Depth-based navigation transfer**: record real depth frames, extend the depth noise model or add image-space randomization, distill the state/range teacher into the depth student (DAgger) to reduce sample cost, then test in a real obstacle course.
6. **Framework**: retrain the goal-reaching tasks with terminal task ends; multi-seed results for the off-policy algorithms and the remaining tasks; a cost critic for constrained RL; pretrained text/vision encoders and fine-tuning of an open VLA on the exported LeRobot data; texture and lighting randomization for RGB policies; exploration and adaptive-control benchmark tasks; recurrent policies for implicit system identification; migration to Isaac Lab 3.x once stable.

11 Reproducibility

- Software: Isaac Sim 5.1.0 (pip), Isaac Lab v2.3.2 (`third_party/IsaacLab`), Python 3.11, torch 2.7.0+cu128, warp-lang 1.11.1, rsl-rl-lib 3.1.2, skrl 1.4.3, onnxruntime 1.20.1 (all pins in `env/constraints.txt`); native installation with `scripts/setup_env.sh`; container `docker/Dockerfile` on the official image `nvcr.io/nvidia/isaac-lab:2.3.2`.
- Every training run stores the exact environment and agent configuration (YAML), command line, git commit and diff summary next to its checkpoints; evaluation JSONs store seeds and overrides.
- Framework checks: `scripts/validation/regression_evals.sh` + `compare_evals.py` (bit-exact reference metrics after core changes), `scripts/validation/examples_check.py` (every extension example in the simulator), `scripts/benchmark/run.py` (suites), `scripts/tools/`

`collect_framework_results.py` (curated results in `docs/results/framework`); LeRobot exports are verified by `scripts/il/export_lerobot.py` in the optional `third_party/lerobot-venv`.

- Validation: `pytest tests/core` (unit tests), `scripts/validation/m{1,2,3,6}_*.py` (milestone acceptance tests), `scripts/rl/evaluate.py` (policy evaluation); tables in this report are generated from `docs/results/*.json` by `scripts/tools/report_tables.py`.
- PX4 SITL: `ros2/px4_sitl/setup_px4_sitl.sh` (PX4 v1.16.2, ROS 2 Jazzy, Gazebo Harmonic, px4_msgs from the same PX4 tree), `scripts/sitl/run_sitl.sh` (one experiment from fresh PX4 parameters to landing), the metric scripts `compare.py`, `compare_tasks.py`, `estimator_stats.py` and `hover_v1_summary.py` in `scripts/sitl/`; the raw recordings (ROS-side `.npz`, bridge step logs, PX4 `.ulg`) are kept in `logs/sitl/`.

References

[1] E. Kaufmann, L. Bauersfeld, A. Loquercio, M. Müller, V. Koltun, D. Scaramuzza. Champion-level drone racing using deep reinforcement learning. *Nature* 620, 2023.

[2] J. Eschmann, D. Albani, G. Loianno. Learning to fly in seconds. *IEEE RA-L*, 2024.

[3] E. Kaufmann, L. Bauersfeld, D. Scaramuzza. A benchmark comparison of learned control policies for agile quadrotor flight. *ICRA*, 2022.

[4] M. Kulkarni, W. Rehberg, K. Alexis. Aerial Gym Simulator: A framework for highly parallelized simulation of aerial robots. *IEEE RA-L*, 2025.

[5] B. Xu et al. OmniDrones: An efficient and flexible platform for reinforcement learning in drone control. *IEEE RA-L*, 2024.

[6] T. Lee, M. Leok, N. H. McClamroch. Geometric tracking control of a quadrotor UAV on SE(3). *IEEE CDC*, 2010.

[7] L. Pinto, M. Andrychowicz, P. Welinder, W. Zaremba, P. Abbeel. Asymmetric actor critic for image-based robot learning. *RSS*, 2018.

[8] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov. Proximal policy optimization algorithms. arXiv:1707.06347, 2017.

[9] T. Haarnoja, A. Zhou, P. Abbeel, S. Levine. Soft actor-critic: off-policy maximum entropy deep reinforcement learning with a stochastic actor. *ICML*, 2018.

[10] S. Fujimoto, H. van Hoof, D. Meger. Addressing function approximation error in actor-critic methods. *ICML*, 2018.

[11] Y. Seo, C. Sferrazza, H. Geng, M. Nauman, Z.-H. Yin, P. Abbeel. FastTD3: simple, fast, and capable reinforcement learning for humanoid control. arXiv, 2025.

[12] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, M. G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. *NeurIPS*, 2021.

[13] K. Black et al. π_0: a vision-language-action flow model for general robot control. *arXiv:2410.24164*, 2024.

[14] A. Stooke, J. Achiam, P. Abbeel. Responsive safety in reinforcement learning by PID Lagrangian methods. *ICML*, 2020.

[15] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, Y. Wu. The surprising effectiveness of PPO in cooperative multi-agent games. *NeurIPS Datasets and Benchmarks*, 2022.

[16] Y. Song, M. Steinweg, E. Kaufmann, D. Scaramuzza. Autonomous drone racing with deep reinforcement learning. *IROS*, 2021.

[17] S. Ross, G. Gordon, D. Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. *AISTATS*, 2011.

[18] P. de Haan, D. Jayaraman, S. Levine. Causal confusion in imitation learning. *NeurIPS*, 2019.

[19] E. Perez, F. Strub, H. de Vries, V. Dumoulin, A. Courville. FiLM: visual reasoning with a general conditioning layer. *AAAI*, 2018.

[20] H. Mania, A. Guy, B. Recht. Simple random search of static linear policies is competitive for reinforcement learning. *NeurIPS*, 2018.

[21] R. Cadene et al. LeRobot: state-of-the-art machine learning for real-world robotics in PyTorch. Software, `https://github.com/huggingface/lerobot`, 2024.

[22] J. Förster. System identification of the Crazyflie 2.0 nano quadrocopter. BSc thesis, ETH Zurich, 2015.